

From Design to Disclosure*

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Abstract

We study the role of hard information in contractual and market settings in which the receiver can flexibly adjust allocations and transfers in response to the sender's disclosure. These settings include monopoly pricing, insurance contracting, and policy negotiations. Across these settings, the sender is worst off if she reveals her type completely because the receiver would then adjust the terms of trade to extract her surplus. Taking this feature as our central departure from the literature, we characterize the entire set of equilibrium payoffs in these disclosure games. We prove an equivalence result: a payoff profile can be approximated by an equilibrium of the disclosure game if and only if it can be achieved through information design. Thus, there exist equilibria in which the sender obtains approximately her commitment payoff.

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1 Introduction

In many strategic interactions, a player can disclose hard information to influence others' actions. An important literature initiated by Grossman (1981) and Milgrom (1981) models such settings as *disclosure* or *persuasion games*. These games feature two core ingredients. First, the informed party—or the sender—possesses hard information that cannot be manipulated: she can share nothing but the truth, although she need not share the entire truth. Second, she cannot precommit to what she discloses; of the utterances that she can make, she chooses the one that results in the most favorable outcome. The party hearing these utterances—the receiver—is no fool. He draws inferences from both what is said and what is left unsaid. Thus, what the sender says and how the receiver responds are jointly determined in equilibrium.

Prior work has shown that combining these ingredients leads to *unraveling*, where the sender discloses all information in the unique equilibrium. The logic for why a fully revealing equilibrium exists is that were the sender to withhold information, the receiver would assume the worst and respond in a way that makes concealment unprofitable. A more powerful force pushes every equilibrium to be fully revealing: in settings modeled by this literature, for any pool of sender types, at least one type strictly prefers to reveal itself rather than remain pooled. The ability to disclose hard information thus creates a *commitment trap*: even if the sender prefers ex ante to withhold information, strategic forces compel her to reveal it all.

While in some applications, the sender may prefer revealing her type to remaining pooled, this property fails in canonical settings. Consider a buyer-seller interaction in which the buyer can disclose information about her value before the seller makes a take-it-or-leave-it (TIOLI) offer. The buyer gains nothing from revealing her true value because doing so enables the seller to extract her full surplus. Similarly, an insuree may disclose some information to an insurer to alleviate adverse selection but disclosing it all would result in a contract that drives her utility down to her outside option.

More generally, in contractual and market settings where the receiver can flexibly adjust the terms of trade based on what is disclosed and then issue a TIOLI offer, full disclosure leaves the sender no better off than her outside option. In such settings, she may prefer to withhold information to counteract the receiver's bargaining power but she cannot commit to doing so. Prior work has not systematically characterized the equilibrium implications of hard information in these settings. Our goal is to provide a unified analysis. Does unraveling fail and, if so, what is the resulting set of equilibrium payoffs?

We study sender-receiver games in which the sender discloses *evidence* about her type to a receiver who then chooses an action, which could be a price offer or a menu of contracts. Like Grossman (1981) and Milgrom (1981), we study rich evidence, where the sender can disclose any subset of types that contains her true type. Our primary departure is in the

payoffs we model: we assume that the sender never strictly gains from fully revealing her type, relative to inducing any other belief. This assumption models the settings above in which full disclosure results in the receiver extracting full surplus. In addition, we impose two standard assumptions. First, every message contains a “worst-case type” whom no type capable of sending that message would like to mimic. Second, the receiver’s payoff satisfies a form of continuity with respect to his actions and beliefs. All three assumptions are satisfied in familiar settings, including monopoly pricing, insurance contracting, and policy negotiations.

Unraveling fails in our setting: while full disclosure is an equilibrium, there also exists one in which the sender fully conceals her type. This multiplicity raises a central question: what is the span of equilibrium payoffs? Generally, characterizing the equilibrium payoff set is infeasible once unraveling fails; existing approaches are constructive and problem-specific. In this game, however, information design can be used to identify all equilibrium payoffs.

To that end, suppose the sender could commit to an information structure that reveals information to the receiver before he chooses his action. Call a payoff profile (u_S^*, u_R^*) *achievable* if some information structure induces it. Every equilibrium payoff of the disclosure game is achievable because any sender strategy can be replicated by an information structure. Our main finding, [Theorem 1](#), establishes the converse.

Main Result. *Every achievable payoff profile can be (approximately) supported by an equilibrium of the disclosure game.*

This result implies that, under our assumptions, there is virtually no gap between information design and voluntary disclosure. Equilibrium outcomes encompass not only full and no revelation but also the entire range of partially revealing outcomes. In particular, the sender’s payoff in some equilibria could be arbitrarily close to that delivered by her optimal information structure. In other words, evidence need not be self-defeating: the sender can not only escape the commitment trap of unraveling but also attain her commitment payoff without having to commit. Yet, the game also has a fully revealing equilibrium that delivers the receiver’s preferred outcome. Thus, the sender’s value for commitment hinges on equilibrium selection.¹

Our finding differs from that of [Grossman \(1981\)](#) and [Milgrom \(1981\)](#) because the receiver can flexibly adjust the terms of trade following disclosure and issue a TIOLI offer. The sender then has an incentive to keep the receiver in the dark so as to counter his bargaining power.

¹Related work that evaluates the commitment assumption in persuasion asks whether the sender accrues her commitment payoff in *some* equilibrium rather than across all equilibria; e.g., [Lipnowski \(2020\)](#), [Kamenica and Lin \(2025\)](#), [Arieli and Stewart \(2025\)](#), and [Dai, Fudenberg, and Pei \(2026\)](#). The work on receiver commitment ([Glazer and Rubinstein, 2006](#); [Sher, 2011](#); [Ben-Porath, Dekel, and Lipman, 2019](#)) takes a similar approach; they compare the receiver’s value from committing to a mechanism to his best equilibrium payoff in the game. One exception is [Hart, Kremer, and Perry \(2017\)](#), who show that a refinement selects the equilibrium that achieves the receiver’s commitment value.

This force is missing in models that restrict the receiver to limited instruments, such as a quantity choice with fixed prices (Milgrom, 1981) or that give all the bargaining power to the sender (Grossman, 1981).

Theorem 1 also makes the methodological point that in settings that satisfy our three assumptions, a modeler can characterize equilibrium payoffs by focusing solely on the receiver’s obedience constraints—exactly as in information design—while disregarding the sender’s incentive constraints altogether. This simplification is substantial: once unraveling fails, tracking both players’ incentive constraints typically makes a full characterization intractable.

We offer some intuition for Theorem 1. One reason to expect a gap is that voluntary disclosure introduces interim incentive constraints absent in information design. Specifically, information design lets the sender commit to splitting a type across messages with different payoffs, whereas in voluntary disclosure she mixes across messages only when they yield the same payoff. With this obstacle in mind, our first step shows that any “finite partitional” segmentation—one in which types are partitioned into finitely many segments, and almost none are split—can be supported as an equilibrium of the disclosure game. Our second step evaluates the restriction to partitional segmentations: we show that any payoff profile achieved by an information structure can be approximated using a finite partitional segmentation. Finally, we illustrate that the three main assumptions are necessary: if any one of them fails, there are settings in which the conclusion does not hold.

Theorem 1 reveals that this game has many equilibria. We find that neither the Intuitive Criterion (Cho and Kreps, 1987) nor robustness to the sender lacking evidence with small probability (Dye, 1985) shrinks the equilibrium payoff set. In the context of monopoly pricing, we find that Hart, Kremer, and Perry (2017)’s truth-leaning refinement selects the efficient frontier; symmetrically, a “concealment-leaning” refinement can support inefficient payoffs.

Our baseline analysis focuses on the standard evidence structure in which the sender may disclose any set of types containing her true type (Grossman and Hart, 1980; Grossman, 1981; Milgrom, 1981). Our motivation is twofold. First, doing so isolates the role of the payoff environment: any departure from the Grossman-Milgrom conclusions stems from the payoffs we model, holding the evidence technology fixed. Second, allowing for this richness implies that any wedges between design and disclosure that we identify do not reflect technological limits that force the sender to communicate coarsely.

However, some applications may call for coarser evidence, others for richer. In an extension, we consider arbitrary evidence structures and identify conditions under which an achievable payoff profile can be supported by an equilibrium of that disclosure game. We then specialize the analysis in two directions. Looking towards coarser evidence, we show that in monopoly pricing, equilibria in interval messages cannot approximate the consumer-optimal

segmentation: high- and low-value consumer types must be able to pool while excluding types with intermediate values. However, if consumers can send messages that are unions of at most K intervals, then equilibria can approximate the payoffs of the consumer-optimal segmentation with a $1/\sqrt{K}$ approximation error. Looking towards richer evidence, we show that allowing disclosure of payoff-irrelevant labels leads to an equivalence between design and disclosure under our first two assumptions alone.

2 Model

A sender (she) shares evidence with a receiver (he) who then chooses an action. The sender privately observes her type θ drawn from the compact set $\Theta \subseteq \mathbb{R}^n$ according to the probability measure F that admits a strictly positive density f with respect to the Lebesgue measure on $\mathbb{R}^n$.² Her type determines what she can say: the sender of type θ chooses a message in $\mathcal{M}(\theta) := \{m \in \mathcal{C} : \theta \in m\}$, where $\mathcal{C}$ denotes the collection of all non-empty closed subsets of Θ . Like Grossman (1981) and Milgrom (1981), we interpret each message as *evidence* in that any statement the sender makes, such as “my type is in m ,” must be true. Observe that $\mathcal{M}(\theta)$ includes the fully revealing message $\{\theta\}$ (available only to type θ), the fully concealing message Θ (available to every type), and a spectrum of messages that reveal some but not all information about the sender’s type.

After receiving the sender’s message, the receiver chooses an action a from a compact metrizable space A . The sender’s payoff $u_S(a, \theta)$ is continuous in (a, θ) , while the receiver’s payoff $u_R(a, \theta)$ is upper semicontinuous in a for each θ .

We now define strategies and equilibria for this game. The sender’s strategy is a function $\rho : \Theta \rightarrow \Delta(\mathcal{C})$ in which the support of $\rho(\theta)$ is a subset of $\mathcal{M}(\theta)$ for every type θ . The receiver’s strategy is a function $\tau : \mathcal{C} \rightarrow A$.³ The receiver’s beliefs about the sender’s type are represented by the belief system $\mu : \mathcal{C} \rightarrow \Delta(\Theta)$. An assessment $((\rho, \tau), \mu)$ is a Perfect Bayesian Equilibrium (henceforth, equilibrium) if the following conditions hold:

- (a) *Given her type, the sender discloses evidence optimally*: for every type θ , $\rho(\theta)$ is supported on $\arg \max_{m \in \mathcal{M}(\theta)} u_S(\tau(m), \theta)$.
- (b) *Given the message, the receiver chooses actions optimally*: for every message m , $\tau(m) \in \arg \max_{a \in A} \int_{\Theta} u_R(a, \theta) d\mu(\theta|m)$.
- (c) *Beliefs respect evidence*: for every message m , the receiver’s beliefs $\mu(m)$ have a support that is a subset of m .

²Section 3.5 considers finitely supported F .

³We endow $\mathcal{C}$ with the Hausdorff metric. Throughout our analysis, for a compact metrizable space X , $\Delta(X)$ denotes the set of (Borel) probability measures on X endowed with the weak* topology. Our analysis allows for mixed actions; then, we interpret A as the set of probability distributions over (pure) actions.

- (d) *Bayes' Rule*: the beliefs μ are obtained from F given ρ using Bayes' rule, i.e., μ is a regular conditional probability system.

Our interest is in market and contractual settings in which the receiver flexibly adjusts the terms of trade based on information that is disclosed: [Example 1](#) below describes the leading case of monopoly pricing and [Example 2](#) considers a more general principal-agent setup.

Example 1 (Monopoly Pricing). Consider the standard monopoly pricing problem with incomplete information, augmented with a disclosure stage: the consumer of a good can disclose evidence about her value θ to a monopolist who then responds with a price offer that the consumer can accept or reject. Here, the consumer's type θ is drawn from $\Theta = [\underline{\theta}, \bar{\theta}]$, where $\underline{\theta} \geq 0$, and action $a \in \mathbb{R}$ represents a price. We write $u_S(a, \theta) = \max\{\theta - a, 0\}$ for the consumer's payoff and $u_R(a, \theta) = a \mathbf{1}_{a \leq \theta}$ for the monopolist's payoff, reflecting that if the consumer accepts the price a , trade happens at that price, and if she rejects, each party obtains 0. ■

Example 2 (Principal-Agent Model). Consider a principal-agent setting in which the type of the agent (sender), θ , is drawn from $\Theta = [\underline{\theta}, \bar{\theta}]$. The principal (receiver) offers a menu of pairs (x, t) , where $x \in X$ is an allocation from a compact space and $t \in \mathbb{R}$ is a transfer from the agent to the principal. The agent selects her preferred option from this menu or an outside option, $x_0 \in X$. The agent's payoff from (x, t) is $v_S(x, \theta) - t$ and the principal's payoff is $v_R(x, \theta) + t$, where $v_S(\cdot, \cdot)$ is continuous and $v_R(\cdot, \theta)$ is upper semicontinuous for each θ .

Our setup captures this interaction by specifying that the principal's action a is a compact subset of $X \times \mathbb{R}$ that includes the outside option, $(x_0, 0)$. The agent selects her preferred option from a , breaking ties in the principal's favor. Letting (x_a, t_a) denote type θ 's optimal choice in a , the resulting payoffs are $u_S(a, \theta) = \max_{(x,t) \in a} \{v_S(x, \theta) - t\}$ and $u_R(a, \theta) = v_R(x_a, \theta) + t_a$. ■

3 When Disclosure Attains Design

3.1 The Information Design Benchmark

We use payoffs that stem from information design as a benchmark: suppose the receiver observes the realization of a Blackwell experiment and then chooses an action. We call a distribution over types $G \in \Delta(\Theta)$ a *belief*. A *segmentation* is a Bayes-plausible distribution $\sigma \in \Delta(\Delta(\Theta))$ over beliefs, and a belief in the support of a segmentation is a *segment*. A segmentation *achieves* a payoff profile (u_S^*, u_R^*) if player i 's ex ante expected payoff from the segmentation, given that the receiver best responds, is u_i^* ; a payoff profile is *achievable* if some segmentation achieves it. For instance, in monopoly pricing ([Example 1](#)), the set of achievable payoffs is the “BBM-triangle” ([Bergemann, Brooks, and Morris, 2015](#)).

Every equilibrium payoff profile of the disclosure game is achievable because every sender strategy induces a segmentation. Our main result pertains to the converse, namely conditions under which every achievable payoff can be approximated by an equilibrium of the disclosure game. We turn to these conditions next.

3.2 The Key Assumptions

We impose three assumptions. The first is that the sender does not benefit from fully disclosing her type; this assumption is our primary departure from the literature and captures settings in which the receiver extracts the sender's surplus if she fully reveals her type. The second assumption, standard in the literature, requires that every message contains a type that no other type prefers to imitate and can thus serve as the root for skeptical beliefs. The third assumption imposes a form of continuity on the receiver's utility function.

To formalize these assumptions, let $a^*(G) := \arg \max_{a \in A} \int u_R(a, \theta) dG(\theta)$ be the receiver's optimal actions given belief G . Let $\underline{a}(\theta)$ denote a receiver-optimal action if the sender's type is known to be θ , where the receiver breaks ties among his optimal actions so as to induce the lowest payoff for the type- θ sender.⁴ These definitions lead to our first assumption.

Assumption 1. *The sender never strictly benefits from fully revealing her type relative to any other belief about her type. Formally, for every type θ , belief G such that the support of G contains θ , and receiver-optimal action $a \in a^*(G)$, $u_S(a, \theta) \geq u_S(\underline{a}(\theta), \theta)$.*

[Assumption 1](#) captures contractual and market settings in which the receiver can flexibly adjust his action and transfers after observing the disclosure. In these settings, full revelation results in the lowest possible payoffs for the sender because the receiver would make a TIOLI offer that fully extracts her surplus. For instance, in monopoly pricing ([Example 1](#)), if the consumer reveals her type θ , the monopolist would charge a price equal to θ , leaving her with no surplus. No belief could make her worse off. The same logic applies in [Example 2](#).⁵

We turn to our second assumption.

Assumption 2. *Every message m contains a worst-case type $\hat{\theta}_m$ such that for every $\theta \in m$, $u_S(\underline{a}(\theta), \theta) \geq u_S(\underline{a}(\hat{\theta}_m), \theta)$. In other words, no type in m would prefer to imitate $\hat{\theta}_m$.*

This assumption, standard in the literature, allows the modeler to deter off-path messages by endowing the receiver with “skeptical beliefs” that concentrate on a single type following

⁴In other words, $\underline{a}(\theta) \in \arg \min_{a \in a^*(\delta_\theta)} u_S(a, \theta)$, where δ_θ is the Dirac measure on θ .

⁵Under complete information, the principal offers a menu with two elements: the outside option with zero transfer and the efficient allocation paired with a transfer that extracts the sender's surplus. That is, the principal's action a is the menu $\{(x_0, 0), (x_\theta^*, t_\theta^*)\}$ where x_0 is the outside option, x_θ^* is an efficient allocation, maximizing $v_S(x, \theta) + v_R(x, \theta)$, and the transfer $t_\theta^* := v_S(x_\theta^*, \theta) - v_S(x_0, \theta)$ leaves the sender just indifferent between accepting and rejecting.

an off-path message. It holds in settings that satisfy a payoff-monotonicity condition (e.g., Grossman, 1981; Milgrom, 1981): when every sender type prefers the receiver to hold higher beliefs, the worst-case type for a message is simply the lowest type capable of sending it. More generally, Seidmann and Winter (1997) and Hagenbach, Koessler, and Perez-Richet (2014) show that a worst-case type exists if preferences satisfy a standard single-crossing condition.

A similar logic applies in our leading examples. For monopoly pricing, the worst-case type for a message m is the highest type that could send it, $\hat{\theta}_m := \max_{\theta \in m} \theta$. No other type that could send message m would wish to mimic $\hat{\theta}_m$ because doing so induces the monopolist to charge a price equal to $\hat{\theta}_m$, yielding the consumer a payoff of 0. More generally, in the principal-agent setting of Example 2, Assumption 2 holds whenever the agent's willingness to pay (WTP) for any alternative x , namely $v_S(x, \theta) - v_S(x_0, \theta)$, weakly increases in θ .⁶

Our final assumption imposes a form of continuity. In many of the settings we study, $u_R(a, \theta)$ is discontinuous in a for each θ . Nevertheless, these games satisfy a form of continuity weaker than the standard notion. The condition below distills that underlying property. Let $U_R(G) := \max_{a \in A} \int u_R(a, \theta) dG(\theta)$ be the receiver's expected payoff when he chooses his optimal action for a belief G . Of his best responses to belief G , let $\underline{a}(G)$ and $\bar{a}(G)$, respectively, denote those that minimize and maximize the sender's payoff: $\underline{a}(G) \in \arg \min_{a \in a^*(G)} \int u_S(a, \theta) dG(\theta)$ and $\bar{a}(G) \in \arg \max_{a \in a^*(G)} \int u_S(a, \theta) dG(\theta)$.

Assumption 3. *The following hold:*

- (a) *The receiver's payoff from choosing an optimal action, $U_R(G)$, is continuous and the set of optimal actions, $a^*(G)$, is upper hemicontinuous.*
- (b) *For every belief G and strictly positive ε and δ , there are*
 - *a belief $\underline{H}$ such that the Radon-Nikodym derivative $\frac{d\underline{H}}{dG} \leq 1 + \varepsilon$, and any best response to $\underline{H}$ is in $B_\delta(\underline{a}(G))$; and*
 - *a belief $\bar{H}$ such that the Radon-Nikodym derivative $\frac{d\bar{H}}{dG} \leq 1 + \varepsilon$, and any best response to $\bar{H}$ is in $B_\delta(\bar{a}(G))$.*

Moreover, the functions that send G to $\underline{H}$ and $\bar{H}$, respectively, are measurable.

To elaborate on (a), let $u_R(a, G) := \int u_R(a, \theta) dG(\theta)$ be the receiver's expected payoff when his belief is G . Were $u_R(a, G)$ continuous in (a, G) , part (a) would follow from Berge's

⁶This increasing-WTP assumption holds in standard contracting environments: if X is partially ordered with x_0 as the lowest alternative, the WTP is increasing if $v_S(x, \theta)$ satisfies increasing differences in (x, θ) . Here, $\hat{\theta}_m := \max_{\theta \in m} \theta$ serves as a worst-case type for message m . Observe that given a point mass belief on $\hat{\theta}_m$, the principal optimally offers a menu that comprises the outside option and an efficient alternative for type $\hat{\theta}_m$ paired with the transfer that extracts full surplus from that type. Because type $\hat{\theta}_m$'s WTP is higher than that of all lower types, every type that could send message m obtains zero surplus from this menu.

maximum theorem. As noted above, $u_R(a, G)$ is *not* continuous in (a, G) in any of our applications. Nevertheless, we show that (a) holds in these settings by applying maximum theorems that require weaker conditions than Berge’s theorem.

Part (b) specifies a form of “selection continuity” for receiver best responses: for every belief G , there exists a “nearby” belief $\underline{H}$ such that every best response to $\underline{H}$ lies close to the best response to G that minimizes the sender’s payoff. Symmetrically, there exists a nearby $\overline{H}$ for which every best response lies close to the best response to G that maximizes the sender’s payoff. A sufficient condition is that any “target” best response to belief G can be “uniquely selected” by perturbing G to a “nearby” H such that all best responses to H are close to the target. This property holds if A is convex and $u_R(a, G)$ is strictly concave in a for every G .

3.3 Main Result

In this setting, a fully revealing equilibrium exists: the sender of type θ sends the message $\{\theta\}$ fully revealing her type, obtaining her complete information payoff; following any off-path message, the receiver assigns probability 1 to a worst-case type. Although the sender obtains a lower payoff than leaving the receiver in the dark (Assumption 1), Assumption 2 guarantees that she cannot profitably deviate to any other message. But alongside this equilibrium, there also exists a fully concealing equilibrium. Now, the sender sends the message Θ regardless of her type and the receiver takes his (ex ante) optimal action on the equilibrium path; following an off-path message, he puts probability 1 on a worst-case type. Given Assumptions 1 and 2, the sender finds neither full revelation nor any partially revealing off-path message profitable.

The existence of these two extreme equilibria raises the question of what else is supportable. Typically, an exhaustive characterization is infeasible once unraveling fails. Here, the entire equilibrium payoff set can be identified, using information design as a tool.

Theorem 1. *Suppose Assumptions 1, 2, and 3 are satisfied. For every achievable payoff profile (u_S^*, u_R^*) and every $\varepsilon > 0$, there is an equilibrium of the disclosure game whose payoffs are within ε of (u_S^*, u_R^*) .*

Theorem 1 establishes a virtual equivalence between the equilibrium and achievable payoff sets. Conceptually, it highlights that the sender can obtain her commitment payoff in some equilibrium of the game. Voluntary disclosure thus need not be a commitment trap. Methodologically, it shows that in settings that satisfy our assumptions, one can find all equilibrium payoffs by focusing solely on the receiver’s obedience constraints—exactly as in information design—while disregarding the sender’s incentive constraints altogether. Thus, it offers a new approach for finding the equilibrium set in monopoly pricing, insurance contracting, and veto bargaining. An additional implication of Theorem 1 is that cheap-talk communication—either

alone or alongside hard information—does not enlarge the equilibrium payoff set. In fact, in all applications that we consider, cheap talk alone effectively results in babbling.

Section 3.4 sketches the proof of Theorem 1, which has two steps. Lemma 1 uses Assumptions 1 and 2 to show that any payoff profile achieved by a “finite partitional segmentation” can be supported as an equilibrium of the disclosure game. The second step of the proof, Lemma 2, uses Assumption 3 to show that for every achievable payoff profile, there exists a nearby one that is achieved by a finite partitional segmentation.⁷ We view Lemma 2 as being of independent interest because it formalizes the appropriate condition under which a persuasion payoff can be approximated without having to randomize or split types. We observe that Assumption 3 holds if the receiver’s action space A is convex and his utility function is continuous and strictly concave.⁸

Our argument for Lemma 2 uses the fact that the distribution of types is atomless; Section 3.5 adapts this argument to some finitely supported priors. Section 3.6 shows that our three assumptions are tight insofar as the conclusion fails if one of them is dropped. Finally, Section 3.7 shows that various standard refinements do not shrink the equilibrium payoff set.

3.4 Proof Sketch

Call a segmentation σ *finite* if its support is a finite set and *partitional* if the set of types that belong to multiple segments has F -measure zero.⁹

Lemma 1. *If Assumptions 1 and 2 are satisfied, then every payoff profile achieved by a finite partitional segmentation can be supported by an equilibrium.*

Proof sketch. Fix a finite partitional segmentation σ and a best response $a : \text{supp } \sigma \rightarrow A$ for the receiver. In the equilibrium we construct, the set of on-path messages is $\{\text{supp } G\}_{G \in \text{supp } \sigma}$. For any on-path message $\text{supp } G$, the receiver updates his belief to G and plays $a(G)$. For any off-path message m , the receiver updates to a point mass belief on the “worst case type” $\hat{\theta}_m$, which exists by Assumption 2, and then plays $\underline{a}(\hat{\theta}_m)$. For any type contained in the support of only one segment in $\text{supp } \sigma$, there is only one on-path message available to her and the sender sends this message. For any type θ in the support of two or more segments in σ , the sender chooses the on-path message available to θ that results in the best possible action given $a(\cdot)$.

We establish that the above constitutes an equilibrium. Observe that the receiver’s beliefs

⁷This payoff profile is nearby in an ex ante sense but may differ in its ex interim payoffs.

⁸Continuity of the utility function guarantees that Assumption 3(a) holds by Berge’s maximum theorem; strict concavity implies that the receiver has a unique best response for every belief G and, because the set of optimal actions $a^*(G)$ is upper hemicontinuous, (b) then holds.

⁹That is, $F(\{\theta : \theta \in \text{supp } G \cap \text{supp } H \text{ for } G \neq H \in \text{supp } \sigma\}) = 0$.

are consistent with Bayes' rule.¹⁰ The receiver plays a best response after every message by construction. Also by construction, no type of the sender has a profitable deviation to an on-path message, and [Assumption 1](#) and [Assumption 2](#) together imply that no type could profitably deviate to an off-path message. ■

Lemma 2. *If [Assumption 3](#) is satisfied, then for every achievable payoff profile (u_S^*, u_R^*) and every $\varepsilon > 0$, there is a finite partitional segmentation that achieves payoffs within ε of (u_S^*, u_R^*) .*

Proof sketch. Fix an achievable payoff profile (u_S^*, u_R^*) , a segmentation σ and a best response achieving this payoff profile, and $\varepsilon > 0$. To illustrate the logic, we assume that the receiver plays $\bar{a}(G)$ for all $G \in \text{supp } \sigma$. [Figure 1](#) depicts the three steps of the argument.



FIGURE 1. The approximation process for [Lemma 2](#).

Because optimal actions may not be lower hemicontinuous in the belief, even small perturbations of the belief can significantly change the receiver's best response and thereby significantly alter the sender's payoff. We sidestep this issue by first perturbing any segment G such that in the perturbed segment, all optimal actions are close to $\bar{a}(G)$: [Assumption 3\(b\)](#) ensures that for every $G \in \text{supp } \sigma$, there is a nearby segment $\bar{H}$ such that any best response to $\bar{H}$ is arbitrarily close to $\bar{a}(G)$. We obtain a new segmentation σ_1 by replacing each $G \in \text{supp } \sigma$ with its corresponding $\bar{H}$.¹¹ By choosing $\bar{H}$ sufficiently close to G , the receiver's payoff under σ_1 is within $\varepsilon/3$ of u_R^* because by [Assumption 3\(a\)](#), the receiver's payoff from choosing an optimal action is continuous in the belief. The sender's payoff under σ_1 is also within $\varepsilon/3$ of u_S^* because any best response to $\bar{H}$ is arbitrarily close to $\bar{a}(G)$ and because the sender's payoff is continuous in the action.

The second step converts σ_1 to a finite segmentation σ_2 by “merging” segments in σ_1 that are sufficiently close to each other into a single segment, which is the average of the aforementioned segments. Because $\Delta(\Theta)$ is compact, the number of resulting “average segments” can be chosen to be finite. Because all best responses to any segment in σ_1 are squished together and the receiver's best-response correspondence is upper hemicontinuous, any best response to an “average segment” in σ_2 is sufficiently close to any best response to any segment in σ_1 that is merged into it. Consequently, the payoffs under σ_2 are within $\varepsilon/3$ of those under σ_1 .

¹⁰Since σ is finite partitional, the set of types that are contained in multiple segments has F -measure zero. Therefore, the behavior of such types does not affect the receiver's beliefs.

¹¹To satisfy Bayes' plausibility, we lower the probability of each segment slightly and create an additional segment.

Our last step identifies a finite partitional segmentation σ_3 that achieves payoffs within $\varepsilon/3$ of those under σ_2 . Loosely speaking, we partition the type space into sufficiently small cubes, and approximate each of the finitely many segments in σ_2 using a collection of such cubes, which is possible because F is absolutely continuous. The statement on payoffs then follows from [Assumption 3\(a\)](#). Therefore, the payoffs under σ_3 are within ε of (u_S^*, u_R^*) . ■

3.5 Finite Types

The proof above uses the fact that the prior is atomless. We find that the same conclusion holds for a finitely supported prior in which no single type has excessive mass. For this analysis, we hold fixed the setting of [Section 2](#), including an ambient type space $\Theta \subseteq \mathbb{R}^n$ that is compact, and consider a prior F whose support is a finite subset of Θ .¹²

Theorem 2. *Suppose Assumptions 1-3 hold. For every $\varepsilon > 0$, there is $\gamma > 0$ such that if F has finite support with $F(\{\theta\}) \leq \gamma$ for every type θ , then for every achievable payoff profile (u_S^*, u_R^*) , there is an equilibrium of the disclosure game whose payoffs are within ε of (u_S^*, u_R^*) .*

The key step of the argument adapts the proof of [Lemma 2](#) to show that once types have sufficiently low mass, any finite segmentation can be approximated (in the appropriate sense) by a partitional one. That no type has excessive mass plays a role because a finite segmentation can split types and a partitional one cannot; were a type to have excessive mass, then some achievable payoff might require substantial splitting across segments, which would make it unattainable by a partitional segmentation. We illustrate this principle below.

Example 3. Consider the setting of [Example 1](#) but suppose the prior is supported on $\{\underline{\theta}, \bar{\theta}\}$ in which $\underline{\theta} > 0$. Suppose the fraction of high-value types ($\bar{\theta}$) is sufficiently high that the price the monopolist would charge based on his prior—i.e., the optimal uniform price—is $\bar{\theta}$, resulting in no buyer surplus. The buyer-optimal segmentation generates buyer surplus by splitting type $\bar{\theta}$ into two segments: one comprising it alone and the other featuring a pool of both types such that the monopolist prices at $\underline{\theta}$ for that pool. Now, some high-value types purchase at a discount. No partitional segmentation could achieve nearby payoffs: either the types separate into segments given by $\{\bar{\theta}\}$ and $\{\underline{\theta}\}$ or they pool into the single unsegmented market $\{\bar{\theta}, \underline{\theta}\}$, and both of these segmentations result in zero buyer surplus. This challenge ports over to the disclosure game: in equilibrium, type $\bar{\theta}$ mixes between messages $\{\bar{\theta}\}$ and $\{\underline{\theta}, \bar{\theta}\}$ only if the two messages result in the same price. Given that the former message results in a price of $\bar{\theta}$, the buyer's equilibrium payoff must be 0. ■

While it may appear restrictive to exclude the binary-type case, [Theorem 2](#) implies that for a sequence of finitely supported distributions converging to that supported on $\{\underline{\theta}, \bar{\theta}\}$ —e.g.,

¹²Observe that whether Assumptions 1-3 hold is independent of the prior.

distributions that put probability mass only on neighborhoods around $\underline{\theta}$ and $\bar{\theta}$ —our conclusion holds along the sequence so long as no single type has excessive mass. The issue therefore arises only at the limit, when the receiver’s prior concentrates on a small number of types.

3.6 What if the Key Assumptions Fail?

Herein, we show that our central conclusion does not hold if any assumption is dropped.

Example 4 ([Assumption 1](#) fails). We consider a stylized version of the [Grossman-Milgrom](#) model. The sender’s type is uniformly distributed on $\Theta = [0, 1]$ and the receiver chooses an action a in $[0, 1]$. The sender has transparent motives in that $u_S(a, \theta)$ does not vary with θ and is strictly increasing and strictly concave in a . The receiver has quadratic-loss payoff $u_R(a, \theta) = -(a - \theta)^2$. [Assumption 1](#) fails in this game: the complete information payoff for the sender of type $\theta = 1$ is higher than her payoff if the receiver’s beliefs equal the prior.

Here, unraveling ensues: the unique equilibrium outcome coincides with full revelation. However, given strict concavity, the best achievable payoff for the sender comes from the receiver being uninformed and choosing action $1/2$. ■

Example 5 ([Assumption 2](#) fails). Suppose $A = \{1, 2\}$, the sender’s type θ is uniformly distributed on $\Theta = [0, 1]$, the receiver’s payoffs are $u_R(1, \theta) = 1 - \theta$ and $u_R(2, \theta) = \theta$, and the sender’s payoffs are $u_S(1, \theta) = \theta$ and $u_S(2, \theta) = 1 - \theta$. [Assumption 2](#) fails in this game, as we show. Observe that the complete-information payoff is θ for types strictly below $\frac{1}{2}$ and $1 - \theta$ for all other types. We argue that the message $m = \Theta$ lacks a worst-case type. If the putative “worst-case” type $\hat{\theta}_m$ were assigned to be strictly below $\frac{1}{2}$, then types above $\frac{1}{2}$ accrue more than their complete-information payoff; were $\hat{\theta}_m$ assigned to be above $\frac{1}{2}$, then types strictly below $\frac{1}{2}$ do better than their complete-information payoff.

This failure is consequential. Given any equilibrium, the sender can deviate to always sending the fully concealing message Θ , which assures her a payoff of $1/2$ regardless of how the receiver responds.¹³ Hence, her equilibrium payoff is at least $1/2$. The payoff profile $(\frac{1}{4}, \frac{3}{4})$ achieved by the fully revealing experiment therefore cannot be approximately supported by an equilibrium of the disclosure game.

Example 6 ([Assumption 3](#) fails). Suppose there are n types, $\{\theta_1, \dots, \theta_n\}$, and $n + 1$ actions, $A = \{0, 1, \dots, n\}$. The sender’s payoff is $u_S(a, \theta) = \mathbf{1}_{a=0}$ and the receiver’s payoff is $u_R(a, \theta) = \mathbf{1}_{a=0} + n\mathbf{1}_{a \neq 0, \theta = \theta_a}$. [Assumption 3\(b\)](#) fails here: every action is optimal under a uniform belief but whenever the belief is not uniform, action 0 is not optimal.

We construct an achievable payoff profile that cannot be approximated in the disclosure game. Take a prior that mixes the uniform belief, with weight α , and a point mass at θ_1 , with

¹³If the receiver chooses action 1 with probability p following the message Θ , then the type- θ sender’s payoff is $p\theta + (1 - p)(1 - \theta)$. Integrating over θ , the sender’s ex ante payoff from this strategy is $1/2$.

weight $(1 - \alpha)$. A segmentation with two segments, one of which is uniform, achieves a payoff of α to the sender. However, in every equilibrium, the sender's payoff must be 0. Observe that following any message $m \neq \Theta$, the receiver would not choose action 0. Thus, the sender accrues a strictly positive payoff only if the receiver played action 0 with positive probability following message $m = \Theta$. But were he to do so, every sender-type would send this message with probability 1, which would make action 0 suboptimal for the receiver. ■

3.7 Refinability of Equilibria

[Theorem 1](#) reveals that the equilibrium payoff set is large. One question is whether refinements reduce the set of supportable payoffs. We find that equilibria survive standard refinements in signaling games and are robust to [Dye](#)-style perturbations in which the sender lacks evidence with small probability. We begin with the Intuitive Criterion of [Cho and Kreps \(1987\)](#).

Theorem 3. *Every equilibrium of the game survives the Intuitive Criterion.*

Here is the idea. An equilibrium fails the Intuitive Criterion only if, after the receiver discards every type for whom an off-path message m is equilibrium-dominated, some surviving type strictly prefers to deviate. None does: each surviving type θ earns at least her complete-information payoff in equilibrium; yet this is also the most the deviation can be forced to yield, because $\underline{a}(\theta)$ is a best response to δ_θ , which is supported on the surviving set. One may wonder whether the stronger refinement of D1 has bite; in the Supplementary Appendix, we show that every equilibrium outcome in the finite-type game survives D1.

We now consider a second refinement, motivated by [Dye \(1985\)](#): we perturb the game so that the sender lacks evidence with probability $\zeta > 0$. Thus, for every type θ , the sender's message space is $\mathcal{M}(\theta)$ with probability $(1 - \zeta)$ and is $\{\Theta\}$ with complementary probability. Denote this perturbed game by Υ^ζ . We say that a payoff profile of the original game Υ^0 is *Dye-supportable* if it is a limit point of payoff profiles of equilibria of Υ^ζ as $\zeta \rightarrow 0$. We show below that all achievable payoffs of the principal-agent setting of [Example 2](#) are Dye-supportable, under one additional assumption.

Assumption 4. *The following hold:*

- (a) *The agent's WTP for every alternative $x \neq x_0$, namely $v_S(x, \theta) - v_S(x_0, \theta)$, strictly increases in her type θ .*
- (b) *The outside option x_0 does not maximize $v_S(x, \bar{\theta}) + v_R(x, \bar{\theta})$.*

We view [Assumption 4](#) to be mild. Part (a) holds whenever X is partially ordered with x_0 as the lowest alternative and $v_S(x, \theta)$ satisfies strict increasing differences. Part (b) reflects

some gains from trade: namely, that the receiver would not choose the outside option if he believed that the sender is type $\bar{\theta}$ with probability 1. We observe that both (a) and (b) are satisfied by the two applications discussed in Section 4. Under Assumptions 3 and 4, we find that the equilibria we consider are robust to the possibility that the sender lacks evidence.

Theorem 4. *Consider a principal-agent setting that satisfies Assumptions 3 and 4. Then every achievable payoff profile (u_S^*, u_R^*) is Dye-supportable.*

Theorems 3 and 4 show that some standard refinements do not shrink the equilibrium payoff set. In specialized settings such as monopoly pricing, we obtain sharper predictions from some other refinements.¹⁴ We turn to these settings next.

4 Applications

4.1 Monopoly Pricing with Interdependent Values

This section generalizes the leading example of monopoly pricing to interdependent values. For this interpretation, we describe the sender as the seller of an object who can provide evidence about its quality to a buyer. Quality θ is drawn according to an absolutely continuous CDF F supported on $\Theta = [\underline{\theta}, \bar{\theta}]$. The seller's value from keeping the object is $c(\theta) \geq 0$, which is strictly increasing and continuous. The buyer's value is $v(\theta)$, which is weakly increasing, continuous, and strictly exceeds $c(\theta)$. Prior to sale, the type- θ seller can send a message $m \in \mathcal{M}(\theta)$. The buyer then sets a price $a \in A = [0, v(\bar{\theta})]$, and the seller accepts or rejects the offer. The seller's payoff is $u_S(a, \theta) = \max\{a, c(\theta)\}$, and the buyer's is $u_R(a, \theta) = (v(\theta) - a)\mathbf{1}_{a \geq c(\theta)}$. Observe that $u_S(\cdot, \cdot)$ is continuous and $u_R(\cdot, \theta)$ is upper semicontinuous for every type θ .¹⁵

The efficient outcome here entails trade with probability 1, resulting in an ex ante surplus of $\mathbb{E}[v(\theta)]$. Relative to that benchmark, this market could suffer from adverse selection (Akerlof, 1970) given that type θ trades at price a only if $c(\theta) \leq a$. Absent disclosure, the buyer's payoff is $\bar{u}_R := \max_{a \in A} F(\{c(\theta) \leq a\})\mathbb{E}[v(\theta) - a \mid c(\theta) \leq a]$, reflecting the lemon's problem.

Kartik and Zhong (2026) characterize the achievable payoff set: all payoff profiles in which total surplus is no more than $\mathbb{E}[v(\theta)]$, the seller's payoff is no less than $\mathbb{E}[c(\theta)]$, and the buyer's payoff is no less than $\bar{u}_R$. All such payoffs can be virtually supported in the disclosure game.

¹⁴We observe that in monopoly pricing (and more broadly in the principal-agent problem), the fully revealing equilibrium fails two rounds of elimination of weakly dominated strategies in the agent-normal form. Observe that for any message m , pricing strictly below the lowest type in m is weakly dominated for the monopolist. After eliminating such prices, it then follows that the message $\{\theta\}$ must result in a price no lower than θ and hence a payoff of 0 for the type- θ sender; by contrast, the message Θ never results in a lower payoff and could result in a higher payoff. Therefore, for any type $\theta > \underline{\theta}$, the message $\{\theta\}$ is weakly dominated.

¹⁵This setting accommodates Example 1, modulo the change of player names, with $v(\theta) = \bar{\theta}$ and $c(\theta) = \bar{\theta} - \theta$.

Proposition 1. *For every payoff profile (u_S^*, u_R^*) with $u_S^* \geq \mathbb{E}[c(\theta)]$, $u_R^* \geq \bar{u}_R$, and $u_S^* + u_R^* \leq \mathbb{E}[v(\theta)]$, and every $\varepsilon > 0$, there is an equilibrium of the disclosure game that supports payoffs within ε of (u_S^*, u_R^*) .*

We prove this result by evaluating our three assumptions; that [Assumptions 1](#) and [2](#) hold is easy to check, and the main part of the proof verifies [Assumption 3](#) using arguments from variational calculus.¹⁶ [Proposition 1](#) implies that the seller can partially disclose evidence to induce nearly fully efficient trade without letting the buyer accrue more than $\bar{u}_R$. Thus, the seller can use evidence to counteract the lemon’s problem while reaping all those gains herself.

But other equilibria could also emerge, including a fully revealing one as well as inefficient ones. We find that the “truth-leaning” refinement proposed by [Hart, Kremer, and Perry \(2017\)](#) selects efficient equilibria. Their approach—invoking the Twainian adage, “When in doubt, tell the truth”—studies limit equilibria of perturbed games in which the sender accrues a tiny gain if she shares the whole truth. Formally, for a function $\varepsilon : \Theta \rightarrow \mathbb{R}_{>0}$, consider the perturbed game Γ^ε in which the seller’s payoff increases by $\varepsilon(\theta)$ when the type is θ and she sends message $\{\theta\}$. An equilibrium $((\rho, \tau), \mu)$ of the original game is *truth-leaning* if there exist (i) a sequence of functions ε^n that converges uniformly to $\mathbf{0}$ (where $\mathbf{0}$ is the zero function), and (ii) a sequence $((\rho^n, \tau^n), \mu^n)$ that converges uniformly to $((\rho, \tau), \mu)$ such that for each $n \in \mathbb{N}$, $((\rho^n, \tau^n), \mu^n)$ is an equilibrium of the perturbed game Γ^{ε^n} .¹⁷

Proposition 2. *The following hold:*

- (a) *The payoff profile of every truth-leaning equilibrium is efficient: (u_S^*, u_R^*) is supported by a truth-leaning equilibrium only if $u_S^* + u_R^* = \mathbb{E}[v(\theta)]$.*
- (b) *For every efficient payoff profile, there is a nearby payoff profile supported by a truth-leaning equilibrium: for every achievable (u_S^*, u_R^*) with $u_S^* + u_R^* = \mathbb{E}[v(\theta)]$ and every $\varepsilon > 0$, there is a truth-leaning equilibrium of the disclosure game that supports payoffs within ε of (u_S^*, u_R^*) .*

By breaking the sender’s indifference in favor of disclosure, the truth-leaning equilibrium engenders efficiency: the key idea for (a) is that inefficiency here arises only if seller types in some segment choose not to trade. But this behavior is untenable in the perturbed game because these seller types would then fully reveal.¹⁸ A subtler intuition underlies (b): we show

¹⁶[Example 3](#) illustrates that this result fails for binary types. However, it holds for any prior with finite support so long as the mass on any single type is not too high.

¹⁷[Hart, Kremer, and Perry \(2017\)](#) also require that in any perturbed game, every type of the sender fully reveal her type with positive probability. This requirement has no bite in our setting.

¹⁸This conclusion is specific to monopoly pricing: in a general principal-agent setting, an inefficient equilibrium could nevertheless give rents to the agent. Truth-leaning would then be less powerful.

that for every finite partitional segmentation achieving payoff $(\tilde{u}_S, \tilde{u}_R)$, there is one on the frontier that achieves the same sender payoff, $\tilde{u}_S$. [Proposition 1](#) then yields the conclusion.

Truth-leaning is one way to break the sender’s indifference; equally, one could envision contexts in which the sender’s payoffs get a small bump from the fully concealing message Θ because, say, she values privacy. A “concealment-leaning” refinement would then be compatible with inefficient payoffs, such as those of the fully concealing equilibrium.

Competition Promotes Disclosure: Suppose now that two or more buyers—each of whom has value function $v(\cdot)$ —compete for the object. We consider a disclosure game in which the seller sends a message m in $\mathcal{M}(\theta)$ publicly to the buyers, each then simultaneously offers a price in $[0, v(\bar{\theta})]$, and the seller accepts at most one offer.

Proposition 3. *In a competitive market with two or more buyers, every equilibrium is payoff equivalent to the fully revealing equilibrium.*

The logic is that Bertrand competition forces each buyer to bid his expected value conditional on all available information. Unraveling ensues: a high type would now break away from any pool whenever doing so raises the market price. In this way, competition promotes disclosure. This finding complements [Milgrom and Roberts \(1986\)](#), who study how competition among multiple senders induces greater disclosure to a single receiver. Here, it is competition among multiple *receivers* that spurs a single sender to divulge all information.

4.2 Insurance Contracting

Should an insurance company be allowed to condition its contracts on an insuree’s disclosures? This question has emerged in regulatory debates on genetic tests ([Golinghorst et al., 2022](#)). One view is that allowing insurers to use such information could reduce asymmetric information, making the market more efficient. An opposing view is that insurees may then feel compelled to disclose all information, transferring all surplus to the insurer.

Herein we augment the standard monopolistic insurance model ([Stiglitz, 1977](#); [Chade and Schlee, 2012](#)) with hard evidence. An insuree has initial wealth $w > 0$, faces a potential loss $\ell \in (0, w)$ with probability θ , and has risk preferences represented by a strictly increasing, continuously differentiable, and strictly concave (Bernoulli) utility function $v : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}$. The probability of loss θ is the insuree’s *type* and her private information. Her outside option is to purchase zero insurance. She can purchase insurance from a risk-neutral insurer whose beliefs about the insuree’s type are given by the absolutely continuous CDF F with density f and support $\Theta := [\underline{\theta}, \bar{\theta}] \subseteq (0, 1)$. Without loss, the insurer chooses a menu of contracts $(x(\theta), t(\theta)) \in \mathbb{R}^2$ for each type θ comprising a premium $t(\theta)$ and an indemnity payment $x(\theta)$

in the event of a loss, subject to incentive and participation constraints. The expected profit from a contract (x, t) chosen by a type- θ insuree is $t - \theta x$.

We append a disclosure stage to this problem. After observing her type θ , the insuree sends a message $m \in \mathcal{M}(\theta)$ to the insurer. The insurer then offers a menu of contracts and the insuree selects one of the contracts or chooses no insurance.

To fit this problem into our framework, we follow [Chade and Schlee \(2012\)](#) and treat the insurer as choosing the insuree’s *indirect utility* function a —the expected utility $a(\theta)$ delivered to each type—rather than a menu of contracts. The underlying menu can essentially be recovered from the slope of a . As a consequence, both payoffs are determined by a : the insuree’s is $u_S(a, \theta) = a(\theta)$ and the insurer’s is a profit $u_R(a, \theta)$. The interaction is thus an instance of our disclosure game in which the receiver chooses a feasible indirect-utility function from a compact action set A .¹⁹ With this reformulation, the following conclusion holds.

Proposition 4. *For every achievable payoff profile (u_S^*, u_R^*) and every $\varepsilon > 0$, there is an equilibrium of the disclosure game that supports payoffs within ε of (u_S^*, u_R^*) .*

We show that our three assumptions hold in this setting: [Assumptions 1](#) and [2](#) follow from the structure of monopolistic insurance, and the insuree’s risk aversion makes the insurer’s payoff *almost* strictly concave, yielding the continuity notion of [Remark 1](#). This result suggests that, with rich evidence, disclosure can generate efficiency gains and can even approximate the insuree-optimal information structure. But the fully revealing equilibrium also exists, enabling the insurer to capture all efficiency gains. Our analysis therefore shifts attention from what evidence can do to the problem of equilibrium selection in this market.²⁰

5 Other Evidence Structures

The baseline model endows the sender with the canonical [Grossman-Milgrom](#) evidence structure, which allows her to disclose any set of types containing her true type. In settings in which unraveling holds, one obtains an evidence-irrelevance conclusion because all equilibria are fully revealing (so long as the sender can fully prove her type) regardless of the fine details of the evidence structure. By contrast, in our model, the equilibrium payoff set may depend on the evidence structure. This section studies that dependence, separating the role of incentives from that of the disclosure technology. [Section 5.1](#) identifies conditions on the disclosure technology under which a target segmentation can be implemented as an equilibrium. Focusing

¹⁹The reformulation is detailed in [Appendix C.5](#).

²⁰An important caveat to this analysis is that it presumes that the insurer’s choice of contracts follows disclosure. Were the insurer able to precommit to the contracts he would offer given each message, he could then effectively force the insuree to disclose all information. In these settings, it would be more important to restrict the kinds of evidence available.

on monopoly pricing, [Section 5.2](#) studies simpler evidence structures and identifies how much complexity is needed to approximate the consumer-optimal market segmentation. [Section 5.3](#) considers evidence richer than in our baseline model, showing how certifiable payoff-irrelevant labels allow every achievable payoff to be supported exactly while dropping the continuity assumption.

5.1 A Generalization of [Lemma 1](#)

Let $\mathcal{M} \subseteq \mathcal{C}$ be a Borel set of closed sets with $\Theta \in \mathcal{M}$. Consider a game identical to that of [Section 2](#) in all respects but one: suppose that every type θ can send messages only in $\mathcal{M}(\theta) := \{m \in \mathcal{M} : \theta \in m\}$. We refer to this game as the $\mathcal{M}$ -restricted disclosure game. Relative to the baseline model, this disclosure game gives each type a smaller set of messages that it can communicate. We define equilibrium for the $\mathcal{M}$ -restricted disclosure game as in [Section 2](#) except we weaken (a) to hold only for an F -measure one set of types.²¹

This technological constraint potentially limits what is feasible in the game. To capture this constraint, we say that a partitional segmentation σ is $\mathcal{M}$ -certifiable if for σ -a.e. segment G , $\text{supp } G$ is in $\mathcal{M}$: in other words, almost all segments in σ could be generated by the messages in $\mathcal{M}$. The following result generalizes [Lemma 1](#) to this setting.

Lemma 3. *Suppose [Assumptions 1](#) and [2](#) hold. Then every payoff profile achieved by a partitional $\mathcal{M}$ -certifiable segmentation can be supported by an equilibrium of the $\mathcal{M}$ -restricted disclosure game.*

[Lemma 3](#) separates the roles of incentives and the evidence technology: given [Assumptions 1](#) and [2](#), a target partitional segmentation can be implemented by an equilibrium so long as (almost) every segment can be certified by the available evidence. This result connects segmentations to the evidence needed to support them. A fully concealing equilibrium always exists because the set Θ can always be certified. A fully revealing equilibrium exists if each type is *completely provable*, i.e., $\{\theta\} \in \mathcal{M}$ for every θ ([Lipman and Seppi, 1995](#)). In a one-dimensional type space, if all *monotone partitions* are $\mathcal{M}$ -certifiable, then every payoff achieved by a monotone segmentation can be supported by an equilibrium of the $\mathcal{M}$ -restricted disclosure game; in fact, the equilibrium would then use only interval messages.²² This implication generalizes [Glode, Opp, and Zhang \(2018\)](#) and [Ali, Lewis, and Vasserman \(2023\)](#),

²¹This weakening changes neither the sender's expected payoff nor the receiver's beliefs and optimal actions but simplifies our proof of [Lemma 3](#) below. We observe that all the other results in this section continue to hold under our original equilibrium definition.

²²A partition is monotone if for each cell, P_i , and triple $\theta < \theta' < \theta''$, θ' lies in P_i whenever both θ and θ'' lie in P_i . We say that an equilibrium uses *interval messages* if, for every type θ , the sender's strategy $\rho(\theta)$ puts probability 1 on some closed interval.

who study interval equilibria, and complements Mensch (2021), Onuchic and Ray (2023), Liu (2024), and Kolotilin, Li, and Zapechelnyuk (2026), who study optimal monotone partitions.

Lemma 3 also recasts our main result, Theorem 1, in terms of the evidence needed for implementation. Since Lemma 2 approximates any achievable payoff by a partitional segmentation, Lemma 3 implies that the approximation is supportable whenever the available evidence can certify the relevant cells. The full Grossman-Milgrom structure guarantees this, but much coarser evidence may certify the relevant one.

5.2 When Does Simpler Evidence Suffice?

We next ask when restricted evidence can support the sender-optimal information-design payoff. We study this question in our leading monopoly-pricing example (Example 1) and use the consumer-optimal segmentation as the benchmark.

One potential evidence structure allows only the extreme messages: $\mathcal{M}(\theta) = \{\{\theta\}, \Theta\}$. Ali, Lewis, and Vasserman (2023, Proposition 1) show that although unraveling fails, no equilibrium improves consumer surplus over the optimal uniform price. The reason is that generating consumer surplus requires partial disclosure, which this structure precludes.

Merely allowing partial disclosure is not enough. Consider the evidence structure in which every interval is a message, echoing the monotone partitions of Section 5.1. Even this structure generally fails to attain the payoffs of the consumer-optimal segmentation. To state this assertion precisely, call an aggregate market *unimprovable* if the monopolist’s optimal uniform price equals the lowest type $\underline{\theta}$. Unimprovable markets are atypical: in most markets, the monopolist prices out low-valuation consumers.

Proposition 5. *There are equilibria in interval messages that approximate the consumer-optimal segmentation if and only if the aggregate market is unimprovable.*

For the “if” direction, the fully concealing equilibrium delivers the consumer-optimal payoffs. For the “only if” direction, the key observation is that an interval message that pools low and high types to induce a price discount is also available to all intermediate types. Those types must then be able to secure, in equilibrium, at least the same discount. The monopolist finds it optimal to serve all of these types at a discount only in an unimprovable market.

What interval evidence misses is the ability to pool low and high value types while preventing intermediate ones from mimicking the pool. In that spirit, we consider equilibria in which (on-path) messages are unions of at most $K \geq 2$ intervals. Such equilibria can approximately support the consumer-optimal segmentation if K is large, with a $\frac{1}{\sqrt{K}}$ -convergence rate.

Proposition 6. *Let (u_S^*, u_R^*) be the consumer-optimal payoff profile. There is an equilibrium of the disclosure game in which every on-path message is a union of at most K intervals and whose payoff profile (u_S^K, u_R^K) satisfies*

$$|u_S^K - u_S^*| + |u_R^K - u_R^*| \leq \frac{C}{\sqrt{K}},$$

where C is independent of K .

This result quantifies the complexity of evidence required to approximate the consumer-optimal segmentation; a union of at most K intervals can get reasonably close.

5.3 Evidence with Payoff-Irrelevant Labels

We next consider evidence richer than in the baseline model: we augment the game with payoff-irrelevant labels. Suppose that in addition to her payoff-relevant type θ , the sender privately observes an independent realization x of a random variable $X \sim U[0, 1]$. Her type is now (θ, x) , and she can send any $m \in \tilde{\mathcal{C}}$ such that $(\theta, x) \in m$, where $\tilde{\mathcal{C}}$ is the collection of all non-empty closed subsets of $\Theta \times [0, 1]$.²³

Proposition 7. *Suppose Θ is finite and [Assumptions 1 and 2](#) hold. Then every achievable payoff profile is supportable.*

This result departs from [Theorem 2](#) in two respects. First, the achievable payoff profile is supported exactly rather than virtually. Second, the labels allow us to drop [Assumption 3](#). The key idea is that this setting allows one to assign different realizations that share the same payoff-relevant type to different segments. With finite Θ , an achievable payoff profile can then be achieved by a finite partitional segmentation (we note that the restriction to finite Θ appears essential). [Lemma 1](#) ensures that it can then be supported by an equilibrium of the disclosure game, using evidence both about the payoff-relevant type and the label.²⁴

²³An enriched state space appears in [Green and Stokey \(2022\)](#), [Kamenica and Gentzkow \(2011, OA Section 3\)](#), and [Gentzkow and Kamenica \(2017\)](#) who represent each signal as a measurable function of an auxiliary random variable. Our approach here fixes an independent auxiliary variable as a primitive of the game and makes it certifiable.

²⁴Our result complements [Kamenica and Gentzkow \(2011, OA Section 2\)](#). They study a disclosure game in which an uninformed sender publicly chooses a Blackwell experiment and can then disclose any set that contains the realization. They show that every equilibrium of this game supports the sender-optimal information structure. Our setting does not feature an uninformed sender choosing a Blackwell experiment, and the equilibria here support all achievable payoffs. [Assumptions 1 and 2](#) are also needed for this result; e.g., the conclusion of [Proposition 7](#) fails in [Examples 4 and 5](#).

6 Discussion and Related Literature

This paper studies voluntary disclosure in contractual settings in which the receiver can flexibly adjust prices, transfers, and allocations in response to the sender’s disclosure and make a TIOLI offer. In these settings, full revelation leaves the sender worst off: once her type is known, the receiver tailors the terms of trade to extract all of her surplus. We take this structural feature as our primary departure from the prior literature. Combined with two standard assumptions, it delivers an equivalence result: the set of equilibrium payoffs in the disclosure game virtually coincides with that achievable through information design. Thus, in settings satisfying our assumptions, a modeler can use information design to find the equilibrium payoff set. This equivalence also suggests that in some contractual settings, the sender can attain her commitment payoff in some equilibrium of the disclosure game.

We now place our findings in the context of prior work. A large literature identifies obstacles to unraveling, including [Dye](#)-evidence structures and costly disclosure; see [Ben-Porath, Dekel, and Lipman \(2025\)](#) for a survey. Our work belongs to the strand that studies payoff environments in which unraveling fails; in this vein, see [Okuno-Fujiwara, Postlewaite, and Suzumura \(1990\)](#), [Seidmann and Winter \(1997\)](#), [Giovannoni and Seidmann \(2007\)](#), [Mathis \(2008\)](#), [Martini \(2018\)](#), and [Titova and Zhang \(2025\)](#). Closer to our work, [Glode, Opp, and Zhang \(2018\)](#), [Pram \(2021\)](#), and [Ali, Lewis, and Vasserman \(2023\)](#) model how evidence can be beneficial in markets and bilateral trade; [Assumption 1](#) holds in their settings and their focus is largely on constructing Pareto improvements.²⁵ Relative to these papers, a distinctive feature of our analysis is that we characterize the entire equilibrium payoff set; [Assumptions 1 to 3](#) isolate the conditions on payoffs that make this tractable.

Our results also connect to work on the role of commitment and randomization in Bayesian persuasion. One strand studies persuasion with limited commitment ([Lipnowski, Ravid, and Shishkin, 2022](#); [Lin and Liu, 2024](#)). Another asks whether persuasion requires randomization: [Malamud and Schrimpf \(2022\)](#), [Arieli et al. \(2023\)](#), and [Zeng \(2023\)](#) provide conditions under which achievable payoffs can be achieved by deterministic segmentations. These conditions fail in our setting. [Arieli and Stewart \(2025\)](#) and [Dai, Fudenberg, and Pei \(2026\)](#) study whether a sender who runs tests and selectively discloses results can achieve her commitment payoff. [Kamenica and Lin \(2025\)](#) compare cheap talk to the sender-optimal information structure; for large action spaces, they show that the fraction of preference profiles for which commitment and randomization are valuable converges to 1. In each of our applications, the action space is large and cheap talk would be ineffectual; hence the disclosure of *hard* rather than *soft* information plays a substantial role.

²⁵Also studying bilateral trade, [Madarasz and Pycia \(2025\)](#) show that information costs can make information choice irrelevant.

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Main Appendix

Appendix A proves Theorem 1 and Appendix B proves Theorem 2. All other proofs are in the Supplementary Appendix.

A Proof of Theorem 1

The proof follows from Lemmas 1 and 2 proven below.

A.1 Proof of Lemma 1

Fix a finite partition segmentation σ . Let (u_S^*, u_R^*) denote the payoff pair that it achieves and $a_\sigma : \text{supp } \sigma \rightarrow A$ be the receiver's best responses that result in these payoffs. Because F has full support, $\bigcup_{G \in \text{supp}(\sigma)} \text{supp } G = \Theta$, as otherwise the segments could not average to F .

Now consider the following sender strategy: If $\theta \in \text{supp}(G)$ and $\theta \notin \text{supp}(H)$ for any $H \in \text{supp}(\sigma)$ such that $H \neq G$, the type- θ sender sends message $\text{supp}(G)$ with probability 1. If $\theta \in \text{supp}(G)$ for multiple $G \in \text{supp}(\sigma)$, type θ sends message $\text{supp } H$ with probability 1, where $H \in \arg \max_{\{G \in \text{supp } \sigma : \theta \in \text{supp } G\}} u_S(a_\sigma(G), \theta)$. Because σ is finite partition, such types are contained in an F -null set and hence do not affect expected payoffs.

The receiver's belief system is such that whenever he observes message $\text{supp}(G)$ for some $G \in \text{supp}(\sigma)$, he updates using Bayes' rule and his new belief is G ; following any other message m , his belief is a point mass at the worst-case type $\hat{\theta}_m$, as defined in Assumption 2. We specify the receiver's strategy as follows: if he observes message $\text{supp}(G)$ for any $G \in \text{supp}(\sigma)$, he chooses action $a_\sigma(G)$ (which is optimal for the receiver given belief G); for any other message m , he chooses action $\underline{a}(\hat{\theta}_m)$.

By construction, the receiver is choosing a best response given his beliefs after any message and beliefs satisfy Bayes' rule whenever possible. Moreover, the sender never wants to deviate from her messaging strategy: by messaging according to the strategy described above, her payoff is at least $u_S(\underline{a}(\theta), \theta)$ by Assumption 1. If she deviates to an off-path message m , her payoff is $u_S(\underline{a}(\hat{\theta}_m), \theta)$, which is lower by Assumption 2. By construction, those types that can send multiple on-path messages choose optimally among feasible on-path messages. Hence, these strategies and beliefs form an equilibrium and this equilibrium induces the segmentation σ and payoffs (u_S^*, u_R^*) . ■

A.2 A Preliminary Step for Lemma 2

Lemma 4. *Let σ be a finite segmentation with $\text{supp}(\sigma) = \{F_1, \dots, F_N\}$. There is a sequence of finite partitional segmentations $\{\sigma^m\}_{m \in \mathbb{N}}$ with $\text{supp}(\sigma^m) = \{F_1^m, \dots, F_N^m\}$ such that, for $i = 1, \dots, N$, $\sigma^m(F_i^m) \rightarrow \sigma(F_i)$ and $F_i^m \rightarrow F_i$.*

Proof. Without loss of generality, suppose $\Theta = [0, 1]^n$. For any $m, k \in \mathbb{N}$, partition Θ into at least m equal-sized cubes; denote this partition by $\mathcal{P}^m$. Partition each $P \in \mathcal{P}^m$ further into at least k equal-sized cubes, and denote the collection of all such small cubes by $\mathcal{Q}$. Choose an assignment of all smaller cubes to $\{1, \dots, N\}$, denoted by $\ell : \mathcal{Q} \rightarrow \{1, \dots, N\}$, to minimize

$$\max_{i \in \{1, \dots, N\}} \sum_{P \in \mathcal{P}^m} \left| F \left(\bigcup_{Q \in \mathcal{Q}, Q \subseteq P, \ell(Q)=i} Q \right) - \sigma(F_i) F_i(P) \right|. \quad (1)$$

Note that this expression goes to zero as $k \rightarrow \infty$. Therefore, for each m , there exists $k(m) \in \mathbb{N}$ such that we can partition each cube into $k(m)$ smaller cubes such that (1) is at most $1/m$. Denote this partition by $\mathcal{Q}^m$.

Define a finite partitional segmentation by setting, for all measurable $B \subseteq \Theta$,

$$F_i^m(B) := \frac{F(B \cap \bigcup_{Q \in \mathcal{Q}^m, \ell(Q)=i} Q)}{F(\bigcup_{Q \in \mathcal{Q}^m, \ell(Q)=i} Q)} \quad \text{and} \quad \sigma^m(F_i^m) := F \left(\bigcup_{Q \in \mathcal{Q}^m, \ell(Q)=i} Q \right).$$

It follows that $\sigma^m(F_i^m) \rightarrow \sigma(F_i)$ and $\sum_{P \in \mathcal{P}^m} |F_i^m(P) - F_i(P)| \rightarrow 0$ as $m \rightarrow \infty$.²⁶ Therefore, for any Lipschitz-continuous function $h : \Theta \rightarrow \mathbb{R}$ and $i \in \{1, \dots, N\}$, $\int h dF_i^m \rightarrow \int h dF_i$. Hence, $F_i^m \rightarrow F_i$. ■

A.3 Proof of Lemma 2

Fix arbitrary $\varepsilon > 0$ and an arbitrary segmentation σ achieving payoffs (u_S^*, u_R^*) .

Step 1: We can assume that the receiver plays $\bar{a}(G)$ for each $G \in \text{supp } \sigma$.

Indeed, the arguments are symmetric if the receiver plays $\underline{a}(G)$ for each $G \in \text{supp } \sigma$, and the conclusion then follows for arbitrary best responses because the sender's payoff is a convex combination of the payoffs achieved from always playing $\bar{a}$ and always playing $\underline{a}$.

Step 2: We choose ε' and δ small enough and define a measurable mapping that maps any

²⁶Indeed, $\sum_{P \in \mathcal{P}^m} |F_i^m(P) - F_i(P)| = \frac{1}{\sigma(F_i)} \sum_{P \in \mathcal{P}^m} \left| \frac{\sigma(F_i)}{F(\bigcup_{Q \in \mathcal{Q}^m, \ell(Q)=i} Q)} F(P \cap \bigcup_{Q \in \mathcal{Q}^m, \ell(Q)=i} Q) - \sigma(F_i) F_i(P) \right| \rightarrow 0$ by definition of ℓ (cf. (1)) and because $\frac{\sigma(F_i)}{F(\bigcup_{Q \in \mathcal{Q}^m, \ell(Q)=i} Q)} \rightarrow 1$.

segment G to a nearby segment $\bar{H}$ such that all best responses in segment $\bar{H}$ are within δ of $\bar{a}(G)$.

Fix $\varepsilon' > 0$ and $\delta > 0$ small enough. Let $t : \Delta(\Theta) \rightarrow \Delta(\Theta)$ be a measurable function that sends any segment G to $t(G)$, where $\frac{dt(G)}{dG} \leq 1 + \varepsilon'$ and any best response given segment $t(G)$ is in $B_\delta(\bar{a}(G))$. [Assumption 3\(b\)](#) ensures that such a function exists.

Step 3: We define σ_1 by considering a (scaled-down) version of the pushforward measure of σ under t and adding a mass point at an extra segment containing the remaining types.

For any (measurable) $Z \subseteq \Delta(\Theta)$, define $\tilde{\sigma}(Z) := \frac{1}{1+\varepsilon'}\sigma(t^{-1}(Z))$ to be a (scaled-down) pushforward of σ under t .

For any (measurable) $E \subseteq \Theta$, $(t(G))(E) = \int_E \frac{dt(G)}{dG} dG \leq (1 + \varepsilon')G(E)$ by the bound on the Radon-Nikodym derivative. Therefore,

$$\left(\int H d\tilde{\sigma}(H) \right) (E) = \left(\frac{1}{1+\varepsilon'} \int t(G) d\sigma(G) \right) (E) \leq \frac{1}{1+\varepsilon'} \int (1+\varepsilon')G(E) d\sigma(G) = F(E)$$

where the first equality follows from the definition of $\tilde{\sigma}$ and the last equality follows since $\int G d\sigma(G) = F$. Intuitively, $\tilde{\sigma}$ does not “exhaust” all types available under the prior (and is not a probability distribution over segments). Therefore, we add a segment G_{extra} that contains all the remaining types: Let $G_{extra} := \frac{F - \int H d\tilde{\sigma}(H)}{1 - \tilde{\sigma}(\Delta(\Theta))}$ and note that $(F - \int H d\tilde{\sigma}(H))(\Theta) = 1 - \frac{1}{1+\varepsilon'} = 1 - \tilde{\sigma}(\Delta(\Theta))$. Hence, $G_{extra} \in \Delta(\Theta)$. Now define $\sigma_1 := \tilde{\sigma} + \delta_{G_{extra}}(1 - \tilde{\sigma}(\Delta(\Theta)))$, where $\delta_{G_{extra}}$ is a Dirac measure at G_{extra} . Simple accounting shows that $\sigma_1 \in \Delta(\Delta(\Theta))$ and $\int G d\sigma_1(G) = F$.

Step 4: We argue that the payoffs under σ_1 are within $\varepsilon/3$ of (u_S^*, u_R^*) .

A fraction $\frac{1}{1+\varepsilon'}$ of types end up in a segment under σ_1 in which any best response for the receiver is within δ of the best response the receiver would have chosen under segmentation σ . Since the sender’s payoff is continuous in the action for each type, we can choose $\varepsilon' > 0$ and $\delta > 0$ in Step 2 small enough such that the expected payoff for the sender is within $\varepsilon/3$ of u_S^* . Similar arguments apply to the receiver’s payoff: We can choose $\varepsilon' > 0$ small enough such that, for any segment G , G and $t(G)$ are close in the Lévy-Prokhorov metric (which metrizes the weak*-topology), and hence the resulting payoffs for the receiver are close by [Assumption 3\(a\)](#). Moreover, by choosing $\varepsilon' > 0$ small enough, the extra segment G_{extra} gets arbitrarily small probability under σ_1 , and hence the receiver’s payoff is within $\varepsilon/3$ of u_R^* .

Step 5: We define a new segmentation σ_2 , which is an approximation of the segmentation σ_1 , such that σ_2 contains finitely many segments and such that payoffs under σ_2 are within $\varepsilon/3$ of the payoffs under σ_1 .

Fix $\delta_2 > 2\delta$. For $\tilde{\sigma}$ -a.e. $G \in \text{supp } \tilde{\sigma} \setminus \{G_{extra}\}$, all best responses lie in a δ -ball (by Step 2) and hence there is an ε_G -ball (in the Lévy-Prokhorov metric) around G such that any best response to any $G' \in B_{\varepsilon_G}(G)$ is within δ_2 of any best response given G (by [Assumption 3\(a\)](#)). Since $\text{supp } \tilde{\sigma}$ is compact, finitely many of these balls, say $\{B_1, \dots, B_m\}$, cover $\text{supp } \tilde{\sigma}$, where $B_i := B_{\varepsilon_{G_i}}(G_i)$. We form a new segmentation σ_2 by merging all segments that lie in a given ball. Formally, σ_2 has at most $m+1$ segments in its support; the first segment is the barycenter of the set B_1 and, for $i > 1$, the i th segment is defined recursively as the barycenter of the set $B_i \setminus \bigcup_{j=1}^{i-1} B_j$ under $\tilde{\sigma}$,

$$H_i := \frac{\int_{B_i \setminus \bigcup_{j=1}^{i-1} B_j} G \, d\tilde{\sigma}(G)}{\tilde{\sigma}(B_i \setminus \bigcup_{j=1}^{i-1} B_j)}$$

whenever $\tilde{\sigma}(B_i \setminus \bigcup_{j=1}^{i-1} B_j) > 0$ and we define $\sigma_2(\{H_i\}) := \tilde{\sigma}(B_i \setminus \bigcup_{j=1}^{i-1} B_j)$ and $\sigma_2(\{G_{extra}\}) := \sigma_1(\{G_{extra}\})$.²⁷ Since open balls in the Lévy-Prokhorov metric are defined by families of affine constraints, they are convex. Therefore, $H_i \in B_i$ and any best response to the merged segment H_i is within $2\delta_2$ of any best response to any $G \in B_i$. By choosing δ_2 small enough, we obtain a segmentation with finite support such that payoffs under σ_2 are within $\varepsilon/3$ of the payoffs under σ_1 .

Step 6: We define a new segmentation σ_3 which is finite partitional and approximates σ_2 . We argue that the payoffs under σ_3 are within $\varepsilon/3$ of the payoffs under σ_2 .

By [Lemma 4](#), for any $\delta' > 0$ we can approximate the segmentation σ_2 by a finite partitional segmentation σ_3 such that for any segment G in σ_2 there is a unique corresponding segment H with $|\sigma_3(H) - \sigma_2(G)| < \delta'$ and $d_P(G, H) < \delta'$, where d_P denotes the Lévy-Prokhorov metric. By [Assumption 3\(a\)](#) we can choose δ' small enough so that the receiver's payoff under segmentation σ_3 is within $\varepsilon/3$ of the payoff under segmentation σ_2 . Similarly, by choosing δ' small enough, any optimal action given belief H is close to any optimal action under G and hence the sender's expected payoff given segmentation σ_3 is within $\varepsilon/3$ of the expected payoff given segmentation σ_2 .

It follows that payoffs under σ_3 are within ε of (u_S^*, u_R^*) . ■

Remark 1. The proof of [Lemma 2](#) still goes through if we replace [Assumption 3\(b\)](#) by the following alternative assumption: *For any $G \in \text{supp } \sigma$ and any $\varepsilon, \delta > 0$, there is a distribution $\underline{H}$ ($\overline{H}$) whose Radon-Nikodym derivative satisfies $\frac{d\underline{H}}{dG} \leq 1 + \varepsilon$ ($\frac{d\overline{H}}{dG} \leq 1 + \varepsilon$) and any best response a' to $\underline{H}$ ($\overline{H}$) is such that $|u_S(a', \theta) - u_S(\underline{a}(G), \theta)| < \delta$ ($|u_S(a', \theta) - u_S(\overline{a}(G), \theta)| < \delta$) for $\underline{H}$ ($\overline{H}$)-almost every θ . Moreover, the functions that send G to $\overline{H}$ and $\underline{H}$ are measurable.*

²⁷For simplicity, we assume $H_i \neq H_j$ for $i \neq j$ and $H_i \neq G_{extra}$. Our arguments apply without this assumption.

Since $\text{supp } \underline{H} \subseteq \text{supp } G$, any action optimal under $\underline{H}$ yields almost the same payoff for types in $\text{supp } \underline{H}$. Thus, by choosing sufficiently small $\varepsilon', \delta > 0$ in Step 2, Step 4 still goes through.

B Proof of Theorem 2

We first prove the following preliminary lemma. Without loss of generality, we assume that Θ is unidimensional.

Lemma 5. *For every $N \in \mathbb{N}$, $\underline{q} > 0$, and $\delta > 0$, there exists $\gamma > 0$ such that the following holds: If F has finite support with $F(\{\theta\}) < \gamma$ for all θ , and σ is a finite segmentation with $\text{supp}(\sigma) = \{F_1, \dots, F_n\}$ where $n \leq N$ and $\sigma(F_i) \geq \underline{q}$ for all i , then there exists a finite partitional segmentation $\bar{\sigma}$ with $\text{supp}(\bar{\sigma}) = \{\bar{F}_1, \dots, \bar{F}_n\}$ such that for each $i = 1, \dots, n$:*

$$|\bar{\sigma}(\bar{F}_i) - \sigma(F_i)| < \delta \quad \text{and} \quad d_P(\bar{F}_i, F_i) < \delta.$$

Proof. Fix $N \in \mathbb{N}$, $\underline{q} > 0$, and $\delta > 0$. Define $\gamma := \frac{q\delta}{N+1}$ and suppose F has finite support with $F(\{\theta\}) < \gamma$ for all θ . Let σ be any segmentation with $n \leq N$ segments satisfying $\sigma(F_i) \geq \underline{q}$ for all i . We construct the finite partitional segmentation $\bar{\sigma}$.

Let $J := \text{supp } F = \{\theta_1, \dots, \theta_M\}$, where for any $s, t \in \{1, \dots, M\}$ with $s < t$, $\theta_s < \theta_t$. Consider an assignment rule $\ell : J \rightarrow \{1, \dots, n\}$ defined as follows: $\ell(\theta_1) = 1$, and for every $s > 1$, define iteratively²⁸

$$\ell(\theta_s) := \min\{i \in \{1, \dots, n\} : F(X_i^{s-1}) \leq \sigma(F_i)F_i(\theta_{s-1})\},$$

where $X_i^{s-1} = \{\theta \in \{\theta_1, \dots, \theta_{s-1}\} : \ell(\theta) = i\}$. For every $i = 1, \dots, n$, define $J_i := \{\theta \in J : \ell(\theta) = i\}$; $\{J_1, \dots, J_n\}$ is a partition of J . Define a finite partitional segmentation $\bar{\sigma}$ with $\text{supp}(\bar{\sigma}) = \{\bar{F}_1, \dots, \bar{F}_n\}$ by setting, for every measurable $B \subseteq \Theta$, $\bar{F}_i(B) := \frac{F(B \cap J_i)}{F(J_i)}$, and $\bar{\sigma}(\bar{F}_i) := F(J_i)$ for each $i = 1, \dots, n$. Since $F(\{\theta\}) < \gamma$ for all $\theta \in J$, we obtain

$$\bar{\sigma}(\bar{F}_i)\bar{F}_i(\theta) \leq \sigma(F_i)F_i(\theta) + \gamma \tag{2}$$

for all $\theta \in \Theta$ and $i = 1, \dots, n$. Because both σ and $\bar{\sigma}$ are segmentations, $\sum_i \bar{\sigma}(\bar{F}_i)\bar{F}_i(\theta) =$

²⁸To ease exposition, we slightly abuse notation by letting $G(\theta) := G([\underline{\theta}, \theta])$ for all $\theta \in \Theta$ for any probability measure G with support on J , where $\underline{\theta} := \inf \Theta$.

$F(\theta) = \sum_i \sigma(F_i) F_i(\theta)$, and hence

$$\sum_{i=1}^n \sigma(F_i) F_i(\theta) = \bar{\sigma}(\bar{F}_k) \bar{F}_k(\theta) + \sum_{i \neq k} \bar{\sigma}(\bar{F}_i) \bar{F}_i(\theta) \leq \bar{\sigma}(\bar{F}_k) \bar{F}_k(\theta) + \sum_{i \neq k} (\sigma(F_i) F_i(\theta) + \gamma)$$

for all $\theta \in J$ and $k = 1, \dots, n$, where the inequality follows from (2). Consequently,

$$\bar{\sigma}(\bar{F}_i) \bar{F}_i(\theta) \geq \sigma(F_i) F_i(\theta) - (n-1)\gamma. \quad (3)$$

Taking $\theta = \theta_M$, inequalities (2) and (3) together imply $\bar{\sigma}(\bar{F}_i) - \sigma(F_i) \leq \gamma$ and $\sigma(F_i) - \bar{\sigma}(\bar{F}_i) \leq (n-1)\gamma$ for each $i = 1, \dots, n$, respectively. Furthermore, dividing by $\sigma(F_i) \geq \underline{q}$ on both sides of (2) and rearranging, we get

$$\bar{F}_i(\theta) - F_i(\theta) \leq \frac{\gamma}{\sigma(F_i)} + \frac{\sigma(F_i) - \bar{\sigma}(\bar{F}_i)}{\sigma(F_i)} \bar{F}_i(\theta) \leq \frac{\gamma}{\underline{q}} + \frac{(n-1)\gamma}{\underline{q}} = \frac{n\gamma}{\underline{q}}$$

for all $\theta \in \Theta$. Similarly, dividing by $\sigma(F_i)$ on both sides of (3), we obtain $F_i(\theta) - \bar{F}_i(\theta) \leq n\gamma/\underline{q}$ for all $\theta \in \Theta$.

Our choice of γ ensures that

$$|\bar{\sigma}(\bar{F}_i) - \sigma(F_i)| \leq (n-1)\gamma \leq (N-1) \cdot \frac{q\delta}{N+1} < \delta,$$

and

$$\sup_{\theta \in \Theta} |\bar{F}_i(\theta) - F_i(\theta)| \leq \frac{n\gamma}{\underline{q}} \leq \frac{N}{\underline{q}} \cdot \frac{q\delta}{N+1} = \frac{N\delta}{N+1} < \delta.$$

By definition of the Lévy-Prokhorov metric, $d_P(\bar{F}_i, F_i) < \delta$. ■

Proof of Theorem 2. Under Assumptions 1 to 3, the proof of Lemma 1 and the first five steps in the proof of Lemma 2 go through without any assumption on the prior F . Fix $\varepsilon > 0$. We show that there exists $\gamma > 0$ such that if F has finite support with $F(\{\theta\}) \leq \gamma$ for all θ , then for every finite segmentation σ , there exists a finite partitional segmentation $\bar{\sigma}$ with payoffs within $\varepsilon/3$ of those under σ .

Step 1: There exists a segmentation $\hat{\sigma}$ with at most K segments, each with probability at least q , and with payoff change from σ of at most $\varepsilon/6$.

Given any segmentation σ with segments $\{F_1, \dots, F_m\}$, we construct a new segmentation as follows: Using Assumption 3 and compactness of $\Delta(\Theta)$, we can cover $\Delta(\Theta)$ by at most K balls $\{B_1, \dots, B_K\}$ such that if we merge any segments in σ that lie in a given ball into a single segment under $\hat{\sigma}$ —using the notion of merging as defined in Step 5 of the proof of Lemma 2—payoffs of the sender and receiver under $\hat{\sigma}$ differ by at most $\varepsilon/12$ from the payoffs

under σ . Note that K depends only on ε . Moreover, we can choose $\underline{q} > 0$ small enough (depending only on ε) such that by merging all segments with probability at most $\underline{q}$ into a single segment, payoffs change by at most $\varepsilon/12$.

Step 2: We apply [Lemma 5](#) to show that there is a finite partitional segmentation $\bar{\sigma}$ with payoff change from $\hat{\sigma}$ of at most $\varepsilon/6$.

By [Lemma 5](#), for every $\delta > 0$ we can find a $\gamma > 0$ such that as long as F has finite support with $F(\{\theta\}) \leq \gamma$, there is a finite partitional segmentation $\bar{\sigma}$ such that to any segment G in $\hat{\sigma}$ there is a unique corresponding segment H with $|\bar{\sigma}(H) - \hat{\sigma}(G)| < \delta$ and $d_P(G, H) < \delta$. By [Assumption 3\(a\)](#) we can choose δ small enough so that the receiver's payoff under segmentation $\bar{\sigma}$ is within $\varepsilon/6$ of the payoff under segmentation $\hat{\sigma}$. Similarly, by choosing δ small enough, any optimal action given belief H is close to any optimal action under G and hence the sender's expected payoff given segmentation $\bar{\sigma}$ is within $\varepsilon/6$ of the expected payoff given segmentation $\hat{\sigma}$. Therefore, the payoffs under $\bar{\sigma}$ are within $\varepsilon/6$ of the payoffs under $\hat{\sigma}$. ■

C Supplementary Appendix

This appendix is organized as follows:

- [Appendix C.1](#) proves that equilibria we consider satisfy the Intuitive Criterion.
- [Appendix C.2](#) proves for the finite-type version of our model that equilibria we consider can be adjusted to satisfy D1.
- [Appendix C.3](#) proves robustness of our main result in principal-agent settings in which the sender lacks evidence with positive probability (“Dye evidence”).
- [Appendix C.4](#) proves results in [Section 4.1](#).
- [Appendix C.5](#) proves [Proposition 4](#).
- [Appendix C.6](#) describes our application to policy negotiations.
- [Appendix C.7](#) proves results in [Section 5](#).

C.1 Proof of Theorem 3

Fix an arbitrary equilibrium and let $v_S^*(\theta)$ denote the interim equilibrium payoff of the type- θ sender. For a belief μ , denote by $BR(\mu) := \arg \max_{a \in A} \int u_R(a, \theta) d\mu(\theta)$ the receiver’s best response correspondence. For any measurable $I \subseteq \Theta$, define $BR(I)$ as the set of responses that are optimal for the receiver under some belief whose support is contained in I : $BR(I) := \bigcup_{\mu: \text{supp } \mu \subseteq I} BR(\mu)$. Finally, for any off-path message m , let

$$S(m) := \left\{ \theta \in m : v_S^*(\theta) > \max_{a \in BR(m)} u_S(a, \theta) \right\};$$

that is, $S(m)$ comprises those types in m whose equilibrium payoff exceeds the highest payoff attainable when the receiver’s belief is concentrated on m .

For any off-path message m with $S(m) \subsetneq m$, and any $\theta \in m \setminus S(m)$, we have

$$v_S^*(\theta) \geq u_S(\underline{a}(\theta), \theta) \geq \min_{a \in BR(m \setminus S(m))} u_S(a, \theta),$$

where the first inequality follows from the fact that every type gets at least her complete-information payoff in any equilibrium, and the second inequality holds because the Dirac measure on θ , δ_θ , satisfies $BR(\delta_\theta) \subseteq BR(m \setminus S(m))$. This implies that the equilibrium $((\rho, \tau), \mu)$ survives the Intuitive Criterion. ■

C.2 D1 in Finite-Type Disclosure Games

In this subsection, we assume Θ is finite and let the prior have full support. We also assume A is the set of all mixed actions available to the receiver. For an arbitrary equilibrium, let $v_S^*(\theta)$ denote the interim equilibrium payoff of the type- θ sender. Since type θ can always send the fully revealing message $\{\theta\}$, every equilibrium satisfies

$$v_S^*(\theta) \geq u_S(\underline{a}(\theta), \theta).$$

We use [Fudenberg and Tirole's \(1991, Definition 11.6\)](#) formulation of D1: Fix an equilibrium e with payoff vector v_S^* , and fix a message m . Let

$$BR(m) := \bigcup_{\nu \in \Delta(\Theta) : \text{supp } \nu \subseteq m} a^*(\nu)$$

be the set of receiver best responses that can be rationalized after m by some evidence-respecting belief. For every $\theta \in m$, define

$$\begin{aligned} D_\theta^+(m) &:= \{a \in BR(m) : u_S(a, \theta) > v_S^*(\theta)\}, \\ D_\theta^0(m) &:= \{a \in BR(m) : u_S(a, \theta) = v_S^*(\theta)\}, \end{aligned}$$

and let

$$W_\theta(m) := D_\theta^+(m) \cup D_\theta^0(m).$$

D1 deletes type $\theta \in m$ after message m if there exists $\theta' \in m$, $\theta' \neq \theta$, such that $W_\theta(m) \subsetneq D_{\theta'}^+(m)$. Let

$$M(m) := \{\theta \in m : \text{there is no } \theta' \in m, \theta' \neq \theta, \text{ with } W_\theta(m) \subsetneq D_{\theta'}^+(m)\}$$

denote the D1-surviving types after message m . We say that an equilibrium outcome survives D1 if the on-path assessment can be completed with off-path beliefs supported on $M(m)$ after every off-path message m .

Theorem 5. *Suppose Assumption 2 holds. Every perfect Bayesian equilibrium outcome survives D1.*

The proof below adjusts our equilibrium construction by changing the off-path beliefs: We show that for any off-path message m there is a set of types $M(m)$ that is not eliminated by D1; [Assumption 2](#) implies that the set $M(m)$ has a worst-case type, and the receiver's belief after observing message m will be a point mass at this type. We show that this belief deters deviations by all types that could have sent m .

Proof. Fix an arbitrary equilibrium $e = ((\rho, \tau), \mu)$. First, every type obtains at least her worst complete-information payoff. Indeed, type θ can send the fully revealing message $\{\theta\}$; after this message, evidence requires the receiver's belief to be δ_θ . Hence

$$v_S^*(\theta) \geq u_S(\tau(\{\theta\}), \theta) \geq u_S(\underline{a}(\theta), \theta).$$

Second, D1 leaves a nonempty set of surviving types after every off-path message. Fix an off-path message m . Define a relation on m by writing $\theta \prec_m \theta'$ if $\theta' \neq \theta$ and $W_\theta(m) \subsetneq D_{\theta'}^+(m)$. Whenever $\theta \prec_m \theta'$, we also have $W_\theta(m) \subsetneq W_{\theta'}(m)$, because $D_{\theta'}^+(m) \subseteq W_{\theta'}(m)$. Since m is finite, any sequence that repeatedly follows $\prec_m$ strictly enlarges the set $W_\theta(m)$ and therefore cannot cycle. Starting from any type, either it is already undeleted or repeatedly following a type that deletes it reaches an undeleted type; hence, $M(m)$ is nonempty.

Finally, a point belief on a worst-case surviving type deters all deviations. Since Θ is finite, the nonempty set $M(m)$ is itself an admissible message. Apply [Assumption 2](#) to $M(m)$ and choose $\hat{\theta} \in M(m)$ such that

$$u_S(\underline{a}(\theta), \theta) \geq u_S(\underline{a}(\hat{\theta}), \theta) \quad \text{for every } \theta \in M(m).$$

Let $a_m := \underline{a}(\hat{\theta})$. Since $\hat{\theta} \in m$ and $a_m \in a^*(\delta_{\hat{\theta}})$, the action a_m belongs to $BR(m)$.

The action a_m deters every type in $M(m)$. For each $\theta \in M(m)$,

$$u_S(a_m, \theta) = u_S(\underline{a}(\hat{\theta}), \theta) \leq u_S(\underline{a}(\theta), \theta) \leq v_S^*(\theta).$$

The same action also deters every deleted type. Suppose $\theta \prec_m \theta'$ and a_m deters θ' . Then $a_m \notin D_{\theta'}^+(m)$. Since $W_\theta(m) \subsetneq D_{\theta'}^+(m)$ and $a_m \in BR(m)$, it follows that $a_m \notin W_\theta(m)$, and therefore $u_S(a_m, \theta) < v_S^*(\theta)$. Now take a chain $\theta_0 \prec_m \theta_1 \prec_m \cdots \prec_m \theta_K$ with $\theta_K \in M(m)$. Since a_m deters θ_K , the preceding implication gives, by backward induction, that a_m deters every type in the chain. Hence a_m deters every type in m .

After off-path message m , set

$$\mu^D(m) = \delta_{\hat{\theta}}, \quad \tau^D(m) = a_m.$$

The belief is supported on $M(m)$ and is therefore D1-admissible. The receiver strategy is a best response to that belief, and no sender type profits from deviating to m .

Repeating this construction after each off-path message, and leaving the original assessment unchanged on path, gives a perfect Bayesian equilibrium assessment with the same outcome as e and with D1-admissible beliefs. ■

C.3 Proof of Theorem 4

As we argued in the main text, both [Assumptions 1](#) and [2](#) hold in this setting. In particular, every message m contains a worst-case type $\hat{\theta}_m := \max_{\theta \in m} \theta$, and the worst-case type of the fully concealing message Θ is $\hat{\theta}_\Theta := \bar{\theta}$.

Fix an achievable payoff profile (u_S^*, u_R^*) . For any $\varepsilon > 0$, [Theorem 1](#) delivers an equilibrium e of the original game Υ^0 whose induced finite partitional segmentation achieves payoffs within ε of (u_S^*, u_R^*) .

For any $\zeta \in (0, 1)$, choose $\xi := \xi(\zeta) > 0$ so that $F((\bar{\theta} - \xi, \bar{\theta}]) = \sqrt{\zeta}$, and define the “top” interval $I_\zeta := (\bar{\theta} - \xi, \bar{\theta}]$. We consider the following strategy profile in the perturbed game Υ^ζ . When the agent has access to all messages in $\mathcal{M}(\theta)$, we prescribe:

- if $\theta \in I_\zeta$, she sends Θ ;
- if $\theta \notin I_\zeta$, she sends the truncated on-path message $\tilde{m}(\theta) := m(\theta) \cap [\underline{\theta}, \bar{\theta} - \xi]$, where $m(\theta)$ is the on-path message sent by type θ in the fixed equilibrium of the original game.

We set the principal’s response after any off-path message m equal to $\underline{a}(\hat{\theta}_m)$; after any on-path message, the principal updates using Bayes’ rule and best responds.

Below, we show that the strategy profile described above is an equilibrium of Υ^ζ for all ζ sufficiently small. We then show that the sequence of associated equilibrium payoff profiles converges as $\zeta \rightarrow 0$ to the payoff profile induced by equilibrium e . Since ε was arbitrary, (u_S^*, u_R^*) is Dye-supportable.²⁹

Step 1: We pin down the principal’s beliefs upon observing Θ and his best response as $\zeta \rightarrow 0$.

Under the strategy defined above, for every Borel subset B of Θ , Bayes’ rule pins down the principal’s posterior belief after observing Θ as

$$\mu_\Theta^\zeta(B) = \frac{\int_B (\zeta + (1 - \zeta)\mathbf{1}_{\theta \in I_\zeta}) dF(\theta)}{\int_\Theta (\zeta + (1 - \zeta)\mathbf{1}_{\theta \in I_\zeta}) dF(\theta)} = \frac{\zeta F(B) + (1 - \zeta)F(B \cap I_\zeta)}{\zeta + (1 - \zeta)F(I_\zeta)}. \quad (4)$$

Since $F(I_\zeta) = \sqrt{\zeta}$, the weight on I_ζ in (4) converges to 1 as $\zeta \rightarrow 0$, and $\xi(\zeta) \rightarrow 0$; hence $\mu_\Theta^\zeta \rightarrow \delta_{\bar{\theta}}$ as $\zeta \rightarrow 0$. Then by [Assumption 3\(a\)](#), for all sufficiently small ζ , every best response to μ_Θ^ζ lies in an arbitrarily small Hausdorff-neighborhood of $a^*(\delta_{\bar{\theta}})$.

Step 2: We show that the strategy profile described above is an equilibrium of Υ^ζ for all sufficiently small ζ .

²⁹Formally, take a sequence of equilibria (e_n) whose payoffs converge to (u_S^*, u_R^*) . For each e_n , there is a sequence of equilibria e_n^k of the perturbed games $\Upsilon^{1/k}$ whose payoffs converge to the payoff of e_n . Then the payoffs of the sequence of equilibria e_n^n of the perturbed games $\Upsilon^{1/n}$ converge to (u_S^*, u_R^*) .

As in [Lemma 1](#), no type profits from deviating to an off-path message, and since σ is finite partitional, almost no type has a profitable deviation to a different on-path message other than Θ . Thus, it suffices to show that no type $\theta \in [\underline{\theta}, \bar{\theta} - \xi]$ finds it profitable to deviate to Θ . Let $X^*(\bar{\theta}) \subseteq X$ denote the set of efficient allocations for type $\bar{\theta}$, i.e.,

$$X^*(\bar{\theta}) := \arg \max_{x \in X} (v_S(x, \bar{\theta}) + v_R(x, \bar{\theta})),$$

and for $x_{\bar{\theta}}^* \in X^*(\bar{\theta})$ let $t(x_{\bar{\theta}}^*) := v_S(x_{\bar{\theta}}^*, \bar{\theta}) - v_S(x_0, \bar{\theta})$. By [Assumption 4](#),

$$v_S(x_{\bar{\theta}}^*, \theta) - v_S(x_0, \theta) < v_S(x_{\bar{\theta}}^*, \bar{\theta}) - v_S(x_0, \bar{\theta}) = t(x_{\bar{\theta}}^*)$$

for any $\theta \in [\underline{\theta}, \bar{\theta} - \xi]$, which implies that $v_S(x_{\bar{\theta}}^*, \theta) - t(x_{\bar{\theta}}^*) < v_S(x_0, \theta)$ for all such types. Moreover, since $X^*(\bar{\theta})$ is compact and $v_S(\cdot, \theta)$ is continuous for every θ , we can define

$$Y(\xi) := \min_{x \in X^*(\bar{\theta})} \left([v_S(x, \bar{\theta}) - v_S(x_0, \bar{\theta})] - [v_S(x, \bar{\theta} - \xi) - v_S(x_0, \bar{\theta} - \xi)] \right) > 0;$$

then for every $x \in X^*(\bar{\theta})$ and every $\theta \in [\underline{\theta}, \bar{\theta} - \xi]$, $v_S(x, \theta) - t(x) \leq v_S(x_0, \theta) - Y(\xi)$.

Now take any menu $a \in a^*(\delta_{\bar{\theta}})$, and let $(x_a, t_a) \in a$ be the option selected by type $\bar{\theta}$ (ties are broken in the principal's favor). Optimality under belief $\delta_{\bar{\theta}}$ implies $x_a \in X^*(\bar{\theta})$ and $v_S(x_a, \bar{\theta}) - t_a = v_S(x_0, \bar{\theta})$, hence $t_a = t(x_a)$. Therefore, any type in $[\underline{\theta}, \bar{\theta} - \xi]$ strictly prefers the outside option to (x_a, t_a) by at least $Y(\xi)$, and the same inequality holds for any other $(x, t) \in a$ because type- $\bar{\theta}$ agent's payoff from (x, t) is at most $\bar{\theta}$'s outside option payoff. Consequently, for every $a \in a^*(\delta_{\bar{\theta}})$ and every $\theta \in [\underline{\theta}, \bar{\theta} - \xi]$, the outside option is the unique optimal choice from a .

By continuity of the agent's payoff in the menu and the definition of $Y(\xi)$, there exists a Hausdorff-neighborhood $\mathcal{E}$ of $a^*(\delta_{\bar{\theta}})$ such that for every menu $a \in \mathcal{E}$ and every $\theta \in [\underline{\theta}, \bar{\theta} - \xi]$, the agent still strictly prefers the outside option and hence obtains exactly $v_S(x_0, \theta)$. Because $a^*(\cdot)$ is upper hemicontinuous by [Assumption 3\(a\)](#) and $\mu_{\Theta}^{\zeta} \rightarrow \delta_{\bar{\theta}}$, for all sufficiently small ζ we have $a^*(\mu_{\Theta}^{\zeta}) \subseteq \mathcal{E}$. Therefore, if the principal best responds to Θ by choosing any $a^{\zeta}(\Theta) \in a^*(\mu_{\Theta}^{\zeta})$, then for every $\theta \in [\underline{\theta}, \bar{\theta} - \xi]$, $u_S(a^{\zeta}(\Theta), \theta) = v_S(x_0, \theta)$. Since [Assumption 1](#) guarantees that each such type's payoff from using the on-path message is at least $v_S(x_0, \theta)$, no type in $[\underline{\theta}, \bar{\theta} - \xi]$ can profitably deviate to Θ . This establishes that the strategy profile defined above is an equilibrium of Υ^{ζ} for all sufficiently small ζ .

Step 3: We show that as $\zeta \rightarrow 0$, the sequence of payoff profiles associated with the equilibria of Υ^{ζ} constructed above converges to the payoff profile of equilibrium e .

We decompose payoffs by considering contributions from types in I_{ζ} and types in $[\underline{\theta}, \bar{\theta} - \xi]$

separately.

For types in I_ζ , since sender and receiver payoffs are bounded and $F(I_\zeta) = \sqrt{\zeta} \rightarrow 0$, the contribution of these types to expected payoffs vanishes as $\zeta \rightarrow 0$.

For types in $[\underline{\theta}, \bar{\theta} - \xi]$: as $\zeta \rightarrow 0$, we have $\xi(\zeta) \rightarrow 0$, so the truncated messages $\tilde{m}(\theta) = m(\theta) \cap [\underline{\theta}, \bar{\theta} - \xi]$ converge to the original messages $m(\theta)$. Consequently, the segment induced by each truncated on-path message converges to the corresponding segment in equilibrium e . By [Assumption 3\(a\)](#), the receiver's expected payoff from best responding is continuous in beliefs, so the receiver's expected payoff from these types converges to that in equilibrium e . By upper hemicontinuity of $a^*(\cdot)$ and compactness of A , any sequence of receiver's best responses to the truncated segments converges to a best response in the original segments. Since $u_S(\cdot, \theta)$ is continuous for each θ and all best responses to a given segment are nearby (by the construction of equilibrium e in [Lemma 2](#) using [Assumption 3\(b\)](#)), the sender's expected payoff from these types also converges to that in equilibrium e .

Consequently, the payoff profiles of the constructed equilibria of Υ^ζ converge to that of equilibrium e as $\zeta \rightarrow 0$. ■

C.4 Proof for Section 4.1 (Monopoly Pricing)

Proof of Proposition 1. Under complete information about type θ , the buyer can guarantee payoff 0 by making an offer below $c(\theta)$ and can obtain payoff $v(\theta) - c(\theta) > 0$ by offering $c(\theta)$. Hence the unique complete-information optimal offer is $c(\theta)$, and the seller's complete-information payoff is $c(\theta)$. Since $u_S(a, \theta) = \max\{a, c(\theta)\} \geq c(\theta)$ for every offer a , [Assumption 1](#) holds. For [Assumption 2](#), for every message m set $\hat{\theta}_m := \min_{\theta \in m} \theta$. Because c is strictly increasing, for every $\theta \in m$ we have $c(\hat{\theta}_m) \leq c(\theta)$, and therefore

$$u_S(\underline{a}(\theta), \theta) = c(\theta) = \max\{c(\hat{\theta}_m), c(\theta)\} = u_S(\underline{a}(\hat{\theta}_m), \theta).$$

Thus no type in m benefits from imitating $\hat{\theta}_m$.

To verify [Assumption 3\(a\)](#), fix an arbitrary G and a sequence $G_n \rightarrow G$. We show that

$$\forall a \in A : \forall a_n \rightarrow a : \quad \limsup_n u_R(a_n, G_n) \leq u_R(a, G) \quad \text{and} \quad (5)$$

$$\forall \varepsilon > 0 : \forall a \in A : \exists a_n \rightarrow a : \quad \liminf_n u_R(a_n, G_n) \geq u_R(a, G) - \varepsilon. \quad (6)$$

This will imply that $u_R(\cdot, G_n)$ Γ -converges to $u_R(\cdot, G)$ (page 288 of [Santambrogio, 2023](#)).³⁰ Propositions 7.4 and 7.5 in [Santambrogio \(2023\)](#) then imply that $U_R(G_n) \rightarrow U_R(G)$ (hence

³⁰More precisely, it will imply that $-u_R(\cdot, G_n)$ Γ -converges to $-u_R(\cdot, G)$. Since we are interested in maximization problems instead of minimization problems, these are the relevant inequalities for us.

$U_R(G)$ is continuous) and the set of optimal actions is upper hemicontinuous.

To verify (5), define

$$\phi(a, \theta) := (v(\theta) - a)\mathbf{1}_{\{a \geq c(\theta)\}}.$$

The function ϕ is bounded and upper semicontinuous on $A \times \Theta$. To see this, fix a convergent sequence $(a_n, \theta_n) \rightarrow (a, \theta)$. If $a < c(\theta)$, then $a_n < c(\theta_n)$ for all sufficiently large n , so $\phi(a_n, \theta_n) = 0$ eventually and hence $\limsup_n \phi(a_n, \theta_n) = 0 = \phi(a, \theta)$. If $a > c(\theta)$, then $a_n \geq c(\theta_n)$ for all sufficiently large n , so continuity of v gives $\lim_n \phi(a_n, \theta_n) = v(\theta) - a = \phi(a, \theta)$. Finally, suppose $a = c(\theta)$. Along any subsequence such that $a_n < c(\theta_n)$, the corresponding payoff is 0; along any subsequence such that $a_n \geq c(\theta_n)$, the corresponding payoff converges to $v(\theta) - a = v(\theta) - c(\theta)$. Since $v(\theta) \geq c(\theta)$, both possible limiting values are weakly below $\phi(a, \theta) = v(\theta) - c(\theta)$. Thus, ϕ is upper semicontinuous. If $a_n \rightarrow a$ and $G_n \rightarrow G$, then $\delta_{a_n} \otimes G_n \rightarrow \delta_a \otimes G$. By the Portmanteau theorem,

$$\limsup_n u_R(a_n, G_n) = \limsup_n \int \phi d(\delta_{a_n} \otimes G_n) \leq \int \phi d(\delta_a \otimes G) = u_R(a, G),$$

which proves (5).

To verify (6), fix $a \in A$. If $a \geq c(\bar{\theta})$, then all types trade at offer a , so $u_R(a, G_n) \rightarrow u_R(a, G)$ by continuity of v , meaning that we can take $a_n = a$. Now suppose $a < c(\bar{\theta})$. Choose a decreasing sequence $\{b_k\}_{k \geq 1}$ with $b_k > a$ for every k and $b_k \rightarrow a$, such that $G(\{\theta : c(\theta) = b_k\}) = 0$ for every k . Such a sequence exists because a probability measure has at most countably many atoms. Because c is strictly increasing, for each fixed k , the function

$$\theta \mapsto (v(\theta) - b_k)\mathbf{1}_{\{b_k \geq c(\theta)\}}$$

is G -almost surely continuous, and hence $u_R(b_k, G_n) \rightarrow u_R(b_k, G)$. Moreover, for every $b > a$,

$$u_R(b, G) - u_R(a, G) = -(b - a)G(\{\theta : c(\theta) \leq a\}) + \int_{\{a < c(\theta) \leq b\}} (v(\theta) - b) dG(\theta) \geq -(b - a),$$

where the inequality follows because $v(\theta) - b \geq c(\theta) - b \geq -(b - a)$ whenever $a < c(\theta) \leq b$. Consequently, $u_R(b_k, G) \geq u_R(a, G) - (b_k - a)$. Choose a strictly increasing sequence $\{N_k\}_{k \geq 1}$ such that, for all $n \geq N_k$,

$$u_R(b_k, G_n) \geq u_R(b_k, G) - \frac{1}{k}.$$

Define $a_n := b_k$ whenever $N_k \leq n < N_{k+1}$, and define a_n arbitrarily for the finitely many remaining n . Then $a_n \rightarrow a$, and $\liminf_n u_R(a_n, G_n) \geq u_R(a, G)$.

It remains to verify Assumption 3(b). Given $\varepsilon, \delta > 0$, we construct two functions $\underline{t} : \Delta(\Theta) \rightarrow \Delta(\Theta)$ and $\bar{t} : \Delta(\Theta) \rightarrow \Delta(\Theta)$ that satisfy Assumption 3(b). Let $C_G := G(c^{-1}(\underline{a}(G))) +$

$(1 - \kappa)(1 - G(c^{-1}(\underline{a}(G)))) = 1 - \kappa + \kappa G(c^{-1}(\underline{a}(G)))$. Define $\underline{t} : \Delta(\Theta) \rightarrow \Delta(\Theta)$ by

$$\underline{t}(G)(x) := \frac{G(x)}{C_G} \mathbf{1}_{\{x \leq c^{-1}(\underline{a}(G))\}} + \frac{G(c^{-1}(\underline{a}(G))) + (1 - \kappa)(G(x) - G(c^{-1}(\underline{a}(G))))}{C_G} \mathbf{1}_{\{x > c^{-1}(\underline{a}(G))\}}.$$

Choose $\kappa > 0$ small enough that $1/(1 - \kappa) \leq 1 + \varepsilon$. Then

$$\frac{d\underline{t}(G)}{dG} \leq 1 + \varepsilon.$$

It is not difficult to see that $\underline{t}$ is measurable.

We next show that for every segment G , $\underline{a}(G)$ is the unique optimal price in segment $\underline{t}(G)$. For any $a' < \underline{a}(G)$,

$$\begin{aligned} \int_{\underline{\theta}}^{c^{-1}(\underline{a})} (v(\theta) - \underline{a}) d\underline{t}(G)(\theta) &= \frac{\int_{\underline{\theta}}^{c^{-1}(\underline{a})} (v(\theta) - \underline{a}) dG(\theta)}{C_G} \\ &> \frac{\int_{\underline{\theta}}^{c^{-1}(a')} (v(\theta) - a') dG(\theta)}{C_G} = \int_{\underline{\theta}}^{c^{-1}(a')} (v(\theta) - a') d\underline{t}(G)(\theta), \end{aligned}$$

where the strict inequality holds because $\underline{a}(G)$ is the lowest optimal offer for belief G . Moreover, for any $a'' > \underline{a}(G)$,

$$\begin{aligned} &\int_{\underline{\theta}}^{c^{-1}(a'')} (v(\theta) - a'') d\underline{t}(G)(\theta) \\ &= \frac{1}{C_G} \left[(1 - \kappa) \int_{\underline{\theta}}^{c^{-1}(a'')} (v(\theta) - a'') dG(\theta) + \kappa \int_{\underline{\theta}}^{c^{-1}(\underline{a})} (v(\theta) - a'') dG(\theta) \right] \\ &\leq \frac{1}{C_G} \left[(1 - \kappa) \int_{\underline{\theta}}^{c^{-1}(\underline{a})} (v(\theta) - \underline{a}) dG(\theta) + \kappa \int_{\underline{\theta}}^{c^{-1}(\underline{a})} (v(\theta) - a'') dG(\theta) \right] \\ &< \frac{1}{C_G} \int_{\underline{\theta}}^{c^{-1}(\underline{a})} (v(\theta) - \underline{a}) dG(\theta) = \int_{\underline{\theta}}^{c^{-1}(\underline{a})} (v(\theta) - \underline{a}) d\underline{t}(G)(\theta), \end{aligned}$$

where the weak inequality holds because $\underline{a}(G)$ is optimal for belief G , and the strict inequality follows since $a'' > \underline{a}(G)$ and, under $v(\theta) > c(\theta)$, any optimal offer yields strictly positive buyer payoff and therefore satisfies $G(c^{-1}(\underline{a}(G))) > 0$.

Define

$$\Delta^+(G) := \max_{a \in [\bar{a}(G), \bar{a}(G) + \delta/2]} u_R(a, G) - \max_{a \geq \bar{a}(G) + \delta} u_R(a, G),$$

with the convention that the second maximum is $-\infty$ if the set is empty. If the set $\{a \in A : a \geq \bar{a}(G) + \delta\}$ is empty, there is nothing to prove for offers above $\bar{a}(G) + \delta$. Otherwise,

$\Delta^+(G) > 0$ because $\bar{a}(G)$ is the highest optimal offer and $u_R(\cdot, G)$ is upper semicontinuous. Let

$$\eta(G) := \frac{1}{2} \min \left\{ \frac{\varepsilon}{1 + \varepsilon}, \frac{\Delta^+(G)}{4v(\bar{\theta}) + \Delta^+(G)}, \frac{U_R(G)}{2v(\bar{\theta}) + U_R(G)} \right\}.$$

Define $\bar{t} : \Delta(\Theta) \rightarrow \Delta(\Theta)$ by trimming the bottom $\eta(G)$ -quantile of G and renormalizing the remaining mass:

$$\bar{t}(G)(x) := \max \left\{ 0, \frac{G(x) - \eta(G)}{1 - \eta(G)} \right\}.$$

Then $\bar{t}$ is measurable and

$$\frac{d\bar{t}(G)}{dG} \leq \frac{1}{1 - \eta(G)} \leq 1 + \varepsilon.$$

We now show that every optimal offer under $\bar{t}(G)$ is within δ of $\bar{a}(G)$. First, for every $a \in A$,

$$|u_R(a, \bar{t}(G)) - u_R(a, G)| \leq \frac{2v(\bar{\theta})\eta(G)}{1 - \eta(G)}. \quad (7)$$

Therefore,

$$\begin{aligned} \max_{a \in [\bar{a}(G), \bar{a}(G) + \delta/2]} u_R(a, \bar{t}(G)) &\geq \max_{a \in [\bar{a}(G), \bar{a}(G) + \delta/2]} u_R(a, G) - \frac{2v(\bar{\theta})\eta(G)}{1 - \eta(G)} \\ &> \max_{a \geq \bar{a}(G) + \delta} u_R(a, G) + \frac{2v(\bar{\theta})\eta(G)}{1 - \eta(G)} \geq \max_{a \geq \bar{a}(G) + \delta} u_R(a, \bar{t}(G)). \end{aligned}$$

The first and third inequalities follow from (7); the strict inequality follows from the definition of $\Delta^+(G)$ and the choice of $\eta(G)$. Thus, no offer $a \geq \bar{a}(G) + \delta$ is optimal under $\bar{t}(G)$.

It remains to rule out offers below $\bar{a}(G)$. Fix $a < \bar{a}(G)$. If $\bar{t}(G)(c^{-1}(a)) = 0$, then $u_R(a, \bar{t}(G)) = 0$. On the other hand, by (7),

$$u_R(\bar{a}(G), \bar{t}(G)) \geq U_R(G) - \frac{2v(\bar{\theta})\eta(G)}{1 - \eta(G)} > 0,$$

so a is not optimal. If instead $\bar{t}(G)(c^{-1}(a)) > 0$, then the lower-tail trimming removes only types with $c(\theta) \leq a$. Hence

$$(1 - \eta(G)) [u_R(\bar{a}(G), \bar{t}(G)) - u_R(a, \bar{t}(G))] = u_R(\bar{a}(G), G) - u_R(a, G) + \eta(G)(\bar{a}(G) - a) > 0,$$

where the weak positivity of the first two terms follows from the optimality of $\bar{a}(G)$ under G , meaning that a is not optimal. Therefore, every optimal offer under $\bar{t}(G)$ belongs to $B_\delta(\bar{a}(G))$. $\blacksquare$

Proof of Proposition 2. To establish (a), it suffices to show that trade must happen with

probability 1 in any truth-leaning equilibrium. In any equilibrium of the perturbed game Γ^ε , trade must happen with probability 1: any type θ that does not trade must fully reveal her type to get payoff $c(\theta) + \varepsilon(\theta) > c(\theta)$, and the unique optimal action for the buyer after receiving message $\{\theta\}$ is $a = c(\theta)$, resulting in trade. By definition, a truth-leaning equilibrium is a limit point of equilibria of Γ^ε , and hence trade also happens with probability 1 in any such equilibrium.

To establish (b), fix $\varepsilon > 0$ and an efficient payoff profile (u_S^*, u_R^*) . By [Proposition 1](#), there exists a payoff profile (u_S^1, u_R^1) that is within $\varepsilon/2$ of (u_S^*, u_R^*) and that is supported by an equilibrium e^* . Moreover, the proof of [Theorem 1](#) shows that we can assume that e^* induces a finite partitional segmentation and each segment has positive probability.

We modify e^* to obtain an efficient equilibrium e^{**} of the unperturbed game that will also be an equilibrium of some perturbed games: In e^{**} , all seller types whose payoff is strictly larger than $c(\theta)$ in e^* send the same message as in e^* ; all types whose payoff is $c(\theta)$ in e^* send the fully revealing messages. The buyer's strategy is as in e^* and beliefs are derived from Bayes' rule whenever possible (with skeptical beliefs after off-path messages).

To verify that e^{**} is an equilibrium of the unperturbed game, note that each seller type gets the same payoff in e^{**} as in e^* , any deviation to an on-path message would yield the same payoff as in e^* , and any deviation to an off-path message is not profitable. Hence, the seller best responds. Moreover, the buyer's actions are still optimal: Consider an on-path message m that is sent with positive probability, and denote the segment and price induced by message m in e^* by G and p_G , respectively. Under the modified strategy of the seller in e^{**} , the segment induced by m is $G(\cdot \mid c(\theta) < p_G)$. Since c is strictly increasing, $F(\{\theta : c(\theta) = p_G\}) = 0$; then since G is absolutely continuous with respect to F , conditioning on $c(\theta) < p_G$ is the same as conditioning on $c(\theta) \leq p_G$. The buyer's payoff from charging $p \leq p_G$ under $G(\cdot \mid c(\theta) < p_G)$ is therefore

$$\tilde{\Pi}(p) = \int_{\{\theta: c(\theta) \leq p\}} (v(\theta) - p) dG(\theta \mid c(\theta) < p_G) = \frac{\int_{\{\theta: c(\theta) \leq p\}} (v(\theta) - p) dG(\theta)}{G(c(\theta) < p_G)}.$$

Since p_G is a maximizer of

$$p \mapsto \int_{\{\theta: c(\theta) \leq p\}} (v(\theta) - p) dG(\theta),$$

it also maximizes $\tilde{\Pi}(p)$ among all $p \leq p_G$. For any $p > p_G$, all seller types in the segment $G(\cdot \mid c(\theta) < p_G)$ trade, and hence

$$\tilde{\Pi}(p) = \tilde{\Pi}(p_G) - (p - p_G) < \tilde{\Pi}(p_G).$$

Therefore, the buyer best responds to m . For any other message, the buyer holds skeptical beliefs and best responds as well.

Denote by (u_S^2, u_R^2) the payoff profile induced by e^{**} . The equilibrium e^{**} is efficient because all types with $c(\theta) < p_G$ trade at price p_G , and all remaining types fully reveal and trade at price $c(\theta)$. To see that (u_S^2, u_R^2) is within ε of (u_S^*, u_R^*) , note that because (u_S^*, u_R^*) and (u_S^2, u_R^2) are efficient and the efficiency frontier has slope -1 ,

$$|u_R^* - u_R^2| = |u_S^* - u_S^2| = |u_S^* - u_S^1| < \frac{\varepsilon}{2}.$$

It remains to argue that the equilibrium e^{**} is truth-leaning. Towards this end, for every $n \in \mathbb{N}$ and for every θ that does not send a fully revealing message in e^{**} , define $\varepsilon^n(\theta) = (p(m) - c(\theta))/2n$, where m is the message sent by type θ in e^{**} and $p(m)$ the resulting price, and for every type θ that sends a fully revealing message in e^{**} , define $\varepsilon^n(\theta) = 1/n$. Then, for each n , $\varepsilon^n(\theta) > 0$ for all θ and e^{**} is an equilibrium of the perturbed game Γ^{ε^n} since no type of the seller has a profitable deviation, and ε^n converges uniformly to $\mathbf{0}$. Therefore, e^{**} constitutes a truth-leaning equilibrium, which completes the proof of (b). ■

Proof of Proposition 3. With two or more buyers engaged in Bertrand competition for the object, every equilibrium involves the highest offer for the object equaling its expected value conditional on disclosure. Consider any belief G of the receiver induced in an equilibrium, and define $\bar{\theta}_G := \max \text{supp } G$. We argue that the expected value of the object under belief G , $\mathbb{E}_G[v(\theta)]$, must satisfy $v(\bar{\theta}_G) - \varepsilon \leq \mathbb{E}_G[v(\theta)] \leq v(\bar{\theta}_G)$ for every $\varepsilon > 0$. The first inequality holds because otherwise some seller type that is inducing belief G would find it profitable to deviate to sending the fully revealing message. The second inequality holds because v is weakly increasing. Given that ε is arbitrary, we obtain $\mathbb{E}_G[v(\theta)] = v(\bar{\theta}_G)$. As v is weakly increasing, this implies that $v(\theta) = v(\bar{\theta}_G)$ for every $\theta \in \text{supp } G$. Therefore, every equilibrium belief is supported on a level set of v . For every type in the support of such a belief, the competitive offer is $\mathbb{E}_G[v(\theta)] = v(\theta)$, which is exactly the offer that would obtain under full disclosure of that type. Hence the induced prices, trade decisions, and payoffs are the same as in the fully revealing equilibrium. Therefore, every equilibrium is payoff equivalent to the fully revealing equilibrium. ■

C.5 Proof of Proposition 4 (Insurance Contracting)

Observe that any incentive-compatible menu of contracts $(x(\theta), t(\theta))$ can be reformulated as a menu $(D(\theta), a(\theta))$, where

$$\begin{aligned} D(\theta) &:= v(w - t(\theta)) - v(w - \ell + x(\theta) - t(\theta)), \\ a(\theta) &:= v(w - t(\theta)) - \theta D(\theta). \end{aligned}$$

If a type- θ insuree accepts the menu, $v(w - t(\theta))$ is her utility when no loss occurs, $D(\theta)$ is the drop in utility when she suffers a loss, and $a(\theta)$ is her *indirect utility*.

We next describe the set of feasible indirect utility functions. By standard arguments, incentive compatibility implies that the indirect utility function $a(\cdot)$ is convex; in this case, $a'(\theta) = -D(\theta)$ almost everywhere. Given the insurer's belief G , any optimal menu satisfies the properties that the participation constraint binds for the lowest type and the utility reduction in the event of a loss, $D(\theta)$, is non-negative and bounded above by $D_0 := v(w) - v(w - \ell)$ for G -almost every θ (Chade and Schlee, 2012, Theorem 1). Observe that D_0 measures the drop in utility from a loss when there is no insurance. Denoting the space of continuous real-valued functions on Θ by $\mathcal{C}(\Theta)$ (equipped with the sup-norm), it is therefore without loss to restrict attention to the following set of indirect utilities:

$$A := \left\{ a \in \mathcal{C}(\Theta) : \begin{array}{l} a(\underline{\theta}) = \underline{\theta}v(w - \ell) + (1 - \underline{\theta})v(w), \\ a \text{ is weakly decreasing, convex and } D_0\text{-Lipschitz} \end{array} \right\}. \quad (8)$$

We now describe the insurer's expected profit as a function of a alone. The insurer's expected profit from a menu (D, a) chosen by a type- θ insuree is

$$u(D, a, \theta) := w - \theta\ell - (1 - \theta)\underbrace{v^{-1}(a(\theta) + \theta D(\theta))}_{\text{no loss}} - \theta\underbrace{v^{-1}(a(\theta) - (1 - \theta)D(\theta))}_{\text{loss}},$$

where $w - \theta\ell$ is type- θ insuree's total wealth in expectation, and the remaining terms are the insurer's expected cost of providing the utilities promised to the insuree. To write the insurer's payoff as a function of a alone, we proceed with the following step. Since $a'(\theta) = -D(\theta)$ almost everywhere, the insurer's payoff is determined by a almost everywhere. At any θ where a is not differentiable, because a is convex and D_0 -Lipschitz, the insurer chooses $D(\theta) \in \partial a(\theta) \cap [0, D_0]$, where $\partial a(\theta)$ is the subdifferential of a at θ . Therefore, we can interpret the indirect utility a as the insurer's action: the insuree's payoff from action $a \in A$ is $u_S(a, \theta) = a(\theta)$, and the insurer's payoff from action a is $u_R(a, \theta) = \max_{D \in \partial a(\theta) \cap [0, D_0]} u(D, a, \theta)$.³¹

³¹Because $\partial a(\theta)$ is closed for every θ and $u(D, a, \theta)$ is continuous, $u_R(a, \theta)$ is well-defined.

We now verify that this formulation is an instance of our general disclosure game.

Lemma 6. *The set A defined in (8) is compact.*

Proof. Let $\{a_n\}_{n \in \mathbb{N}}$ be a sequence in A . Since a_n is uniformly bounded and D_0 -Lipschitz continuous for each n , Arzela-Ascoli's theorem implies there is a subsequence, also denoted by $\{a_n\}$, that converges in the supremum-norm to some a . Clearly, $a \in A$. ■

Lemma 7. *For all θ , $u_S(a, \theta)$ is continuous in a . Also, $u_R(a, G)$ is upper semicontinuous in (a, G) .*

Proof. If $a_n \rightarrow a$ then $u_S(a_n, \theta) = a_n(\theta) \rightarrow a(\theta) = u_S(a, \theta)$ and the first claim follows.

For the second claim, we show first that $u_R(a, \theta)$ is upper semicontinuous in (a, θ) . Note that if $a_n \rightarrow a$, $\theta_n \rightarrow \theta$, and $\varepsilon > 0$ then $\partial a_n(\theta_n) \subseteq B_\varepsilon(\partial a(\theta))$ for all n large enough (Theorem D.6.2.7 in [Hiriart-Urruty and Lemaréchal, 2004](#)). Hence, $(a, \theta) \mapsto \partial a(\theta)$ is upper hemicontinuous. Since $u(D, a, \theta)$ is continuous in (D, a, θ) , a maximum theorem (see Lemma 17.30 in [Aliprantis and Border, 2006](#)) implies that $u_R(a, \theta)$ is upper semicontinuous in (a, θ) .

Now consider $a_n \rightarrow a$ and $G_n \rightarrow G$. By Theorem 25.6 in [Billingsley \(1995\)](#) there are Y_n and Y on a common probability space $(\Omega, \mathcal{F}, P)$ with distributions G_n and G such that $Y_n(\omega) \rightarrow Y(\omega)$ for all ω . Then

$$\begin{aligned} \limsup_{n \rightarrow \infty} \int u_R(a_n, \theta) dG_n(\theta) &= \limsup_{n \rightarrow \infty} \int u_R(a_n, Y_n(\omega)) dP(\omega) \\ &\leq \int u_R(a, Y(\omega)) dP(\omega) = \int u_R(a, \theta) dG(\theta) \end{aligned}$$

where the equalities follow from a change of variables and the inequality follows from (reverse) Fatou's lemma and the fact that $u_R(a, \theta)$ is upper semicontinuous in (a, θ) . We conclude that $u_R(a, G)$ is upper semicontinuous in (a, G) . ■

Lemma 8. *Given $a \in A$, $G \in \Delta(\Theta)$, and $\varepsilon > 0$ there is $\tilde{a} \in A$ that is continuously differentiable such that $\|a - \tilde{a}\|_\infty < \varepsilon/2$ and $|u_R(\tilde{a}, G) - u_R(a, G)| < \varepsilon/2$.*

Proof. Fix $a \in A$; let E' denote the set of θ 's at which a is not differentiable. Because E' is countable, we can find a finite set $E'' = \{\theta_1, \dots, \theta_M\} \subseteq E'$ such that $G(E' \setminus E'') < \varepsilon/(12w)$, where w is an upper bound for $|u_R(a, \theta) - u_R(b, \theta)|$ for any $a, b \in A$. Let $D^*(\theta) \in \arg \max_{D \in \partial a(\theta) \cap [0, D_0]} u(D, a, \theta)$; and for each $i = 1, \dots, M$, let ℓ_i denote the supporting hyperplane of a at θ_i with slope $D^*(\theta_i)$.

For any $n \in \mathbb{N}$, we can find a piecewise affine function $\tilde{a}_n$ such that $\tilde{a}_n \in A$ and $\|\tilde{a}_n - a\|_\infty < 1/(9n)$. Define $\hat{a}_n := \max\{\tilde{a}_n - 1/(9n), \ell_1, \dots, \ell_M\} + 1/(9n)$; it can be shown that $\|\hat{a}_n - \tilde{a}_n\|_\infty < 1/(9n)$, and $\hat{a}_n \in A$ for n large enough. We can approximate $\hat{a}_n$ by a differentiable function

a_n such that $\|a_n - \hat{a}_n\|_\infty \leq 1/(9n)$ and $|a'_n(\theta) - D^*(\theta)| \leq 1/(9n)$ for all $\theta \in E''$ (Rockafellar and Wets, 2009, Theorem 2.26).

Since a_n is convex and differentiable, $a_n \rightarrow a$ implies that $a'_n(\theta) \rightarrow a'(\theta)$ for all $\theta \notin E'$. By Egoroff's theorem, there exists $E \subseteq \Theta \setminus E'$ with $G(E) < \varepsilon/(12w)$ such that $a'_n \rightarrow a'$ uniformly on $\Theta \setminus (E' \cup E)$. Because $|a'_n(\theta) - D^*(\theta)| < 1/(9n)$ for all $\theta \in E''$, $|u_R(a_n, \theta) - u_R(a, \theta)| < \varepsilon/3$ for any $\theta \notin (E' \setminus E'') \cup E$ and all n large enough. Then since $G(E' \setminus E'') < \varepsilon/(12w)$ and $G(E) < \varepsilon/(12w)$, $|u_R(a_n, G) - u_R(a, G)| < \varepsilon/2$ for large enough n . ■

Lemma 9. *The receiver's optimal payoff is continuous in G and the receiver's optimal actions are upper hemicontinuous in G .*

Proof. Because $u_R(a, G)$ is upper semicontinuous (Lemma 7), the optimal payoff $U_R(G) := \max_{a \in A} u_R(a, G)$ is upper semicontinuous by Lemma 17.30 in Aliprantis and Border (2006).

We argue that $U_R(G)$ is also lower semicontinuous: Consider $G_n \rightarrow G$ and suppose towards contradiction there is $\varepsilon > 0$ such that $\liminf_n U_R(G_n) + \varepsilon < U_R(G)$. By Lemma 8, there is a continuously differentiable $\tilde{a} \in A$ such that $u_R(\tilde{a}, G) + \varepsilon/2 > U_R(G)$. Then $u_R(\tilde{a}, \theta)$ is continuous in θ , and therefore $\int u_R(\tilde{a}, \theta) dG_n(\theta) \rightarrow \int u_R(\tilde{a}, \theta) dG(\theta)$ (since we use the weak* topology). Hence, for all n large enough, $u_R(\tilde{a}, G_n) + \varepsilon/2 \geq u_R(\tilde{a}, G)$. This implies $u_R(\tilde{a}, G_n) + \varepsilon \geq U_R(G)$, a contradiction.

Finally, we show that $a^*(G) = \arg \max_{a \in A} u_R(a, G)$ is upper hemicontinuous: Consider sequences $G_n \rightarrow G$ and $a_n \in a^*(G_n)$ such that $a_n \rightarrow a$, and suppose towards contradiction that $a \notin a^*(G)$. Because $a \in A$ this implies $U_R(G) > u_R(a, G)$. Because $u_R(a, G)$ is upper semicontinuous (Lemma 7),

$$U_R(G) > u_R(a, G) \geq \limsup_{n \rightarrow \infty} u_R(a_n, G_n) = \limsup_{n \rightarrow \infty} U_R(G_n).$$

This contradicts lower semicontinuity of U_R . ■

Proof of Proposition 4. Observe that for each type θ , the sender's *no-insurance payoff* is $\theta v(w - \ell) + (1 - \theta)v(w)$. Assumption 1 holds because if the insuree is known to be of type θ , the insurer will offer a contract that extracts all surplus leaving the insuree with her no-insurance payoff. Given that the no-insurance payoff is strictly decreasing in θ , for any $m \in \mathcal{C}$, the worst-case type is $\max_{\theta \in m} \theta$, which verifies Assumption 2. Assumption 3(a) follows from Lemma 9. Instead of verifying Assumption 3(b), we verify its alternative in Remark 1. Because $u_R(a, \theta)$ is strictly concave in a for each θ ,³² for any $G \in \Delta(\Theta)$, any $a', a'' \in a^*(G)$ satisfy $a' = a''$ G -almost everywhere. The alternative assumption holds by setting $t(G) = G$ and $\underline{H} = \overline{H} = t(G)$. ■

³²Take $\lambda \in (0, 1)$, any $a', a'' \in A$ with $a' \neq a''$, and any $D' \in \arg \max_{D \in \partial a'(\theta) \cap [0, D_0]} u(a', D, \theta)$ and

C.6 Bargaining Over Policies

This section applies our analysis to models of policy negotiations with incomplete information that build on [Romer and Rosenthal \(1978\)](#). A policy $a \in \mathbb{R}$ is determined by the proposer and the vetoer. The proposer's payoff from policy a , $u(a)$, is strictly increasing in a . The vetoer's payoff, $v(a, \theta)$, is continuous, strictly single-peaked in a with a unique maximizer θ , and symmetric around the maximizer; we call the vetoer's ideal policy θ her *type*. Her type is her private information, and is drawn according to an absolutely continuous CDF F on $\Theta = [\underline{\theta}, \bar{\theta}]$ with $\bar{\theta} < \infty$. For simplicity, we assume $\underline{\theta} \geq 0$. Once the proposer proposes a policy a , the vetoer can accept or reject. If she accepts, the proposed policy prevails; if she rejects, then the status-quo policy $a_Q = 0$ is preserved. Given the proposer's payoffs, he never proposes any $a < 0$. Restricting attention to $a \geq 0$, the vetoer accepts if and only if $a \leq 2\theta$. Therefore, we assume that proposals lie in $[0, 2\bar{\theta}]$.

We augment this game with a disclosure stage, with the following timing. The vetoer first observes her type θ and sends a message $m \in \mathcal{M}(\theta)$ to the proposer. The proposer then proposes a policy a . The type- θ vetoer's payoff is given by $u_S(a, \theta) = \max\{v(a, \theta), v(0, \theta)\}$, and the proposer's payoff is given by $u_R(a, \theta) = (u(a) - u(0))\mathbf{1}_{a \leq 2\theta}$. For every $\theta \in \Theta$, $u_S(\cdot, \theta)$ is continuous, and $u_R(\cdot, \theta)$ is upper semicontinuous.

We note that this setting does not feature transferable utility and, unlike monopoly pricing, the sender and receiver may have aligned preferences that favor a higher action to a_Q . Nevertheless, the complete-information payoff is the lowest for a sender-type θ because the receiver would then propose action 2θ , which results in the same payoff as the status quo.

In this setting, the implications of information design have not been characterized; nevertheless, the equivalence between design and disclosure holds.

Proposition 8. *For every achievable payoff profile (u_S^*, u_R^*) and every $\varepsilon > 0$, there is an equilibrium of the disclosure game that supports payoffs within ε of (u_S^*, u_R^*) .*

Proof. We verify [Assumptions 1 to 3](#). Because $\underline{a}(\theta) = 2\theta$ and v is symmetric around θ , $v(\underline{a}(\theta), \theta) = v(0, \theta)$. Consequently, $u_S(\underline{a}(\theta), \theta) = v(0, \theta)$. For every $\theta \in \Theta$, $u_S(a, \theta)$ is bounded below by $v(0, \theta)$, and hence [Assumption 1](#) must hold. [Assumption 2](#) also holds because for

$D'' \in \arg \max_{D \in \partial a''(\theta) \cap [0, D_0]} u(a'', D, \theta)$. Then

$$\begin{aligned} \lambda u_R(a', \theta) + (1 - \lambda) u_R(a'', \theta) &< w - \theta \ell - (1 - \theta) v^{-1}((\lambda a'(\theta) + (1 - \lambda) a''(\theta)) + \theta(\lambda D' + (1 - \lambda) D'')) - \\ &\quad \theta v^{-1}((\lambda a'(\theta) + (1 - \lambda) a''(\theta)) - (1 - \theta)(\lambda D' + (1 - \lambda) D'')) \\ &\leq u_R(\lambda a' + (1 - \lambda) a'', \theta), \end{aligned}$$

where the first inequality holds because v^{-1} is strictly convex (since v is strictly concave), and the second inequality follows from the fact that $\lambda D' + (1 - \lambda) D'' \in \partial(\lambda a' + (1 - \lambda) a'')(\theta) = \lambda \partial a'(\theta) + (1 - \lambda) \partial a''(\theta)$ for each θ (Theorem D.4.1.1 in [Hiriart-Urruty and Lemaréchal, 2004](#)).

every message $m \in \mathcal{C}$, one can set $\hat{\theta}_m := \max_{\theta \in m} \theta$. Finally, by re-writing the proposer's payoff as $u_R(b, \theta) = w(b)\mathbf{1}_{b \leq \theta}$, where $b = a/2$ and $w(b) := u(2b) - u(0)$, we can show that [Assumption 3](#) holds by adapting the argument in the proof of [Proposition 1](#). ■

C.7 Proofs for Section 5

We use the following disintegration fact, which follows from the existence of regular conditional distributions on compact metric spaces.

Fact 1. *Let σ be partitional and suppose $\text{supp } G \in \mathcal{M}$ for σ -a.e. segment G . Then there are an F -null set $N \subseteq \Theta$, a measurable map $\phi : \Theta \rightarrow \Delta(\Theta)$, a set $\mathcal{M}_\sigma \subseteq \mathcal{M}$, and a map $\psi : \mathcal{M}_\sigma \rightarrow \Delta(\Theta)$ such that:*

- (i) $\theta \in \text{supp } \phi(\theta)$ for every $\theta \notin N$;
- (ii) the F -law of $\phi(\theta)$ is σ ;
- (iii) $\text{supp } \phi(\theta) \in \mathcal{M}_\sigma$ and $\psi(\text{supp } \phi(\theta)) = \phi(\theta)$ for every $\theta \notin N$; and
- (iv) the regular conditional distribution of θ given the message $\text{supp } \phi(\theta)$ is $\psi(\text{supp } \phi(\theta))$, F -a.e.

We apply this fact by using ϕ as a function that maps types to segments; $\mathcal{M}_\sigma$ will correspond to on-path messages; and ψ will map on-path messages to beliefs of the receiver.

Proof. Let $P \in \Delta(\Delta(\Theta) \times \Theta)$ be the joint law of (G, θ) obtained by drawing $G \sim \sigma$ and then $\theta \sim G$; that is,

$$P(A \times B) = \int_A G(B) \sigma(dG)$$

for Borel $A \subseteq \Delta(\Theta)$ and $B \subseteq \Theta$. Since σ is a segmentation, the Θ -marginal of P is F , while the $\Delta(\Theta)$ -marginal of P is σ . Moreover, by construction, the conditional law of θ given G is G .

Since Θ , and hence $\Delta(\Theta)$, are compact metric spaces, P admits a regular conditional distribution $\theta \mapsto \nu_\theta \in \Delta(\Delta(\Theta))$ of G given θ ([Kallenberg, 2002](#), Theorem 6.3). We claim ν_θ is a Dirac measure for F -a.e. θ . Under P we have $\theta \in \text{supp } G$ for P -a.e. (G, θ) and $G \in \text{supp } \sigma$ for P -a.e. (G, θ) . Hence for F -a.e. θ , ν_θ is concentrated on $\{G \in \text{supp } \sigma : \theta \in \text{supp } G\}$. By partitionality, we can choose a Borel F -null set E such that no $\theta \notin E$ belongs to the supports of two distinct elements of $\text{supp } \sigma$. Enlarging E if necessary, ν_θ is a Dirac measure for every $\theta \notin E$. Let $\phi(\theta)$ be its atom, choosing ϕ measurably and defining it arbitrarily on E .

Now define $\Gamma := \{G \in \text{supp } \sigma : G(E) = 0 \text{ and } \text{supp } G \in \mathcal{M}\}$. Then $\sigma(\Gamma) = 1$, because $\int G(E) \sigma(dG) = F(E) = 0$ and $\text{supp } G \in \mathcal{M}$ for σ -a.e. G by hypothesis. Set $N := E \cup \{\theta : \phi(\theta) \notin \Gamma\}$. Since the F -law of $\phi(\theta)$ is σ , $F(N) = 0$. Thus, for every $\theta \notin N$, $\theta \in \text{supp } \phi(\theta)$, proving (i) and (ii), and the conditional law of θ given $\phi(\theta)$ equals $\phi(\theta)$.

The support map $G \mapsto \text{supp } G$ is one-to-one on Γ : if two distinct such segments shared a support S , then since $G(S) = 1$ and $G(E) = 0$ there is a point $\theta \in S \setminus E$ lying in the supports of two distinct segments of $\text{supp } \sigma$, contradicting partitionality. Let $\mathcal{M}_\sigma$ be the image of Γ under the support map, and define $\psi : \mathcal{M}_\sigma \rightarrow \Delta(\Theta)$ by $\psi(\text{supp } G) = G$. Since conditioning on $\text{supp } \phi(\theta)$ coincides with conditioning on $\phi(\theta)$, (iii) and (iv) follow. $\blacksquare$

Proof of Lemma 3. Fix a partitional segmentation σ for which $a_\sigma : \text{supp}(\sigma) \rightarrow A$ denotes a best response for each segment that results in payoff profile (u_S, u_R) . Let $\mathcal{M}_\sigma$, ψ , ϕ , and N be as in Fact 1 and write $m_\theta := \text{supp } \phi(\theta)$ for $\theta \notin N$.

Strategies and beliefs. For $\theta \notin N$ let $\rho(\theta) = \delta_{m_\theta}$; this is feasible since $m_\theta \in \mathcal{M}$ and $\theta \in m_\theta$. For $\theta \in N$, choose any feasible message, for instance Θ . Set

$$\mu(m) = \begin{cases} \psi(m), & m \in \mathcal{M}_\sigma, \\ \delta_{\hat{\theta}_m}, & m \in \mathcal{M} \setminus \mathcal{M}_\sigma, \end{cases} \quad \tau(m) = \begin{cases} a_\sigma(\psi(m)), & m \in \mathcal{M}_\sigma, \\ \underline{a}(\hat{\theta}_m), & m \in \mathcal{M} \setminus \mathcal{M}_\sigma. \end{cases}$$

We verify this is an equilibrium that induces the right payoffs:

- (i) For $m \in \mathcal{M}_\sigma$, $\text{supp } \psi(m) = m$, so $\text{supp } \mu(m) \subseteq m$. For $m \notin \mathcal{M}_\sigma$, $\text{supp } \mu(m) = \{\hat{\theta}_m\} \subseteq m$ by Assumption 2. Hence, (c) holds.
- (ii) Under (F, ρ) the realized message is m_θ for F -a.e. θ . By Fact 1, the conditional law of θ given m_θ is $\psi(m_\theta)$, so μ restricted to $\mathcal{M}_\sigma$ is a regular conditional distribution on the realized-message distribution. Messages in $\mathcal{M} \setminus \mathcal{M}_\sigma$ are sent with probability zero, so any evidence-respecting belief is Bayes-consistent there. Hence, (d) holds. Moreover, the induced distribution of posterior beliefs is σ .
- (iii) *Receiver optimality.* For $m \in \mathcal{M}_\sigma$, $\tau(m) = a_\sigma(\psi(m)) \in a^*(\psi(m)) = a^*(\mu(m))$. For $m \in \mathcal{M} \setminus \mathcal{M}_\sigma$, $\underline{a}(\hat{\theta}_m) \in a^*(\delta_{\hat{\theta}_m}) = a^*(\mu(m))$. Hence, (b) holds.
- (iv) *Sender optimality.* Fix $\theta \notin N$. The feasible on-path message is m_θ : if another $m \in \mathcal{M}_\sigma$ contained θ , then θ would lie in the supports of two distinct segments, contradicting the assumption that σ is partitional. Sending m_θ yields $u_S(a_\sigma(\phi(\theta)), \theta)$, and since $\theta \in \text{supp } \phi(\theta)$ and $a_\sigma(\phi(\theta)) \in a^*(\phi(\theta))$, Assumption 1 gives

$$u_S(a_\sigma(\phi(\theta)), \theta) \geq u_S(\underline{a}(\theta), \theta).$$

Sending any off-path $m \in \mathcal{M} \setminus \mathcal{M}_\sigma$ that contains θ yields $u_S(\underline{a}(\hat{\theta}_m), \theta)$, and by Assumption 2,

$$u_S(\underline{a}(\theta), \theta) \geq u_S(\underline{a}(\hat{\theta}_m), \theta).$$

Thus m_θ is optimal among all feasible messages. Hence, (a) holds.

Thus $((\rho, \tau), \mu)$ is an equilibrium replicating σ . Since the receiver plays $a_\sigma(\phi(\theta))$ after F -a.e. type's message, the ex-ante payoffs coincide with those generated by σ . ■

Proof of Proposition 5. If the aggregate market is unimprovable, the fully concealing equilibrium—in which every type sends the message $[\underline{\theta}, \bar{\theta}]$ —supports the payoffs of the consumer-optimal segmentation.

For the converse, suppose the aggregate market is improvable and let $p^* > \underline{\theta}$ denote the lowest uniform monopoly price for the aggregate market. Towards a contradiction, suppose that there is an interval-message equilibrium that yields ex ante consumer payoffs of at least $w^* - \pi^* - 1/n$ for every $n \in \mathbb{N}$. The monopolist's payoff in the n^{th} equilibrium is therefore at most $\pi^* + 1/n$ and the welfare loss from exclusion is at most $1/n$.

Consequently, for any $m > 0$ and all n sufficiently large, in the segmentation induced by the n^{th} equilibrium, there is a set of segments $\mathcal{G}_n^m$ whose probability is at least $1 - 1/m$ and such that each segment $G \in \mathcal{G}_n^m$ satisfies: (i) the monopolist's optimal profit in segment G exceeds the profit from charging p^* by at most $1/m$, and (ii) at most a fraction $1/m$ of types in segment G is excluded. Property (i) implies that for all n sufficiently large, each $G \in \mathcal{G}_n^m$ contains types in $[p^*, p^* + 1/m]$. Property (ii) implies that for n sufficiently large, the equilibrium price in each $G \in \mathcal{G}_n^m$ is at most $G^{-1}(1/m)$. It follows that for m and n large enough, $\mathcal{G}_n^m$ contains a segment with equilibrium price close to $\underline{\theta}$.

In equilibrium, each type sends whichever available message yields the lowest price, provided this price is below its valuation. Consequently, for any $\varepsilon > 0$, by properties (i) and (ii) it holds that for all m and n sufficiently large, types in $[\underline{\theta} + \varepsilon, p^*]$ will send messages that result in the same price and at least one of these messages contains a type that is at least $p^* + 1/m$ (otherwise, uniformly lowering the price from p^* in all these segments would increase profit, contradicting that p^* is close to optimal in all segments). But this implies that all types $\theta \in [p^*, p^* + 1/m]$ will in equilibrium send a message resulting in a price close to $\underline{\theta}$. By property (i), all segments in $\mathcal{G}_n^m$ must therefore result in this low price. But as $m, n \rightarrow \infty$, the probability of $\mathcal{G}_n^m$ converges to 1, and because $\underline{\theta}$ is not an optimal uniform price, equilibrium profits fall strictly below π^* , yielding a contradiction. ■

Lemma 10. For all $G, H \in \Delta(\Theta)$,

$$|U_R(G) - U_R(H)| \leq (1 + \bar{\theta})d_P(G, H),$$

where d_P denotes the Lévy-Prokhorov metric.

Proof. Fix $\varepsilon > d_P(G, H)$ and a price $a \in [0, \bar{\theta}]$. Since $\text{supp } H \subseteq [\underline{\theta}, \bar{\theta}]$, the ε -neighborhood of

$[a, \bar{\theta}]$, $[a - \varepsilon, \bar{\theta} + \varepsilon]$, satisfies $[a - \varepsilon, \bar{\theta} + \varepsilon] \cap \text{supp } H \subseteq [a - \varepsilon, \bar{\theta}]$. By the definition of d_P ,

$$G([a, \bar{\theta}]) \leq H([a - \varepsilon, \bar{\theta}]) + \varepsilon.$$

Hence

$$u_R(a, G) \leq (a - \varepsilon)H([a - \varepsilon, \bar{\theta}]) + (1 + \bar{\theta})\varepsilon \leq U_R(H) + (1 + \bar{\theta})\varepsilon.$$

Taking the supremum over $a \in [0, \bar{\theta}]$ gives $U_R(G) \leq U_R(H) + (1 + \bar{\theta})\varepsilon$; reversing the roles of G and H yields the reverse inequality, and letting $\varepsilon \downarrow d_P(G, H)$ gives the claim. $\blacksquare$

For a segment G , write $p^{\min}(G) := \min \text{supp } G$ for the lowest price in its support. Define the top truncation G^η of G by

$$G^\eta(x) := \frac{\min\{G(x), 1 - \eta\}}{1 - \eta}, \quad \eta \in (0, 1/2).$$

Equivalently, G^η deletes the top η -quantile of G and renormalizes the remaining mass.

Lemma 11. *Suppose $\underline{\theta} > 0$. There are constants $C_0, c_0 > 0$, where C_0 depends only on $\bar{\theta}$ and c_0 depends only on $\underline{\theta}$ and $\bar{\theta}$, such that the following holds for every G with $p^{\min}(G) \in a^*(G)$. Parts 1 and 2 do not require $\underline{\theta} > 0$; only the stability bound in Part 3 does.*

1. $G^\eta \ll G$, $dG^\eta/dG \leq (1 - \eta)^{-1}$, and $d_P(G^\eta, G) \leq \eta/(1 - \eta)$.
2. $p^{\min}(G)$ is the unique optimal price under G^η .
3. If $d_P(H, G^\eta) \leq \delta \leq \min\{\eta/4, c_0\}$, then

$$a^*(H) \subseteq \left[p^{\min}(G) - C_0 \frac{\delta}{\eta}, p^{\min}(G) + C_0 \frac{\delta}{\eta} \right].$$

Proof. Parts 1 and 2 are the mirror-image of the bottom-truncation construction from the proof of [Proposition 1](#). The map defined there, $\bar{t}(G)$, is the bottom-truncation analogue of G^η ; that proof therefore gives $dG^\eta/dG \leq (1 - \eta)^{-1}$ and shows that no price strictly above the lowest optimal price of G is optimal under G^η . Since $p^{\min}(G) = \min \text{supp } G \in a^*(G)$, the lowest optimal price is $p^{\min}(G)$, and any lower price sells to the whole market and so earns strictly less than $p^{\min}(G)$; hence $p^{\min}(G)$ is the unique optimal price under G^η . Finally G^η and G differ in total variation by η , so $d_P(G^\eta, G) \leq \|G^\eta - G\|_{TV} = \eta \leq \eta/(1 - \eta)$.

It remains to prove the stability bound. Let $p := p^{\min}(G)$, and fix $\varepsilon \in (d_P(H, G^\eta), 2\delta)$. Since $G^\eta([p, \bar{\theta}]) = 1$,

$$U_R(H) \geq (p - \varepsilon)(1 - \varepsilon) \geq p - (1 + \bar{\theta})\varepsilon.$$

Therefore no price $a \leq p - z$ is optimal if $z > (1 + \bar{\theta})\varepsilon$.

Now take $a \geq p + z$, with $z > 2\varepsilon$, and put $y := a - \varepsilon$. The definition of d_P gives

$$H([a, \bar{\theta}]) \leq G^\eta([y, \bar{\theta}]) + \varepsilon.$$

For $y > p$, optimality of p under G gives $yG([y, \bar{\theta}]) \leq p$. Since top truncation removes the upper η -tail,

$$G^\eta([y, \bar{\theta}]) = \frac{(G([y, \bar{\theta}]) - \eta)_+}{1 - \eta},$$

with the usual left-limit convention at atoms. Hence

$$u_R(y, G^\eta) \leq \left(p - \frac{\eta(y - p)}{1 - \eta} \right)_+.$$

If the positive part is zero, then $G^\eta([y, \bar{\theta}]) = 0$, so $u_R(a, H) \leq \bar{\theta}\varepsilon$, while $U_R(H) \geq \underline{\theta} - (1 + \bar{\theta})\varepsilon$; choosing $c_0 \leq \underline{\theta}/[2(1 + 2\bar{\theta})]$ rules out optimality of a . Otherwise,

$$u_R(a, H) \leq u_R(y, G^\eta) + (1 + \bar{\theta})\varepsilon \leq p - \frac{\eta(z - \varepsilon)}{1 - \eta} + (1 + \bar{\theta})\varepsilon.$$

Comparing this with $U_R(H) \geq p - (1 + \bar{\theta})\varepsilon$, price a is not optimal whenever

$$\frac{\eta(z - \varepsilon)}{1 - \eta} > 2(1 + \bar{\theta})\varepsilon.$$

Taking $C_0 = 2(1 + \bar{\theta})$, no such price is optimal once $z > C_0\varepsilon/\eta$. Letting $\varepsilon \downarrow d_P(H, G^\eta)$ proves the claim. ■

Lemma 12. *Let $\{(q_i, G_i)\}_{i=1}^N$ be a finite segmentation of F . There is a partition $\{P_i^K\}_{i=1}^N$ of Θ , up to endpoints, such that each P_i^K is a union of at most K intervals, $F(P_i^K) = q_i$, and, writing $H_i^K := F(\cdot \mid P_i^K)$,*

$$d_P(H_i^K, G_i) \leq \frac{\bar{\theta} - \underline{\theta}}{K} \quad \text{for each } i.$$

Proof. Partition Θ into K equal-length cells $\mathcal{I}_K = \{I_1, \dots, I_K\}$, each of length $\Delta := (\bar{\theta} - \underline{\theta})/K$. Let $\mu_i := q_i G_i$. Since $\sum_i \mu_i = F$, atomlessness of F allows us to split each cell I_r into subintervals $\{P_{ir}\}_i$, up to endpoints, satisfying $F(P_{ir}) = \mu_i(I_r)$. Set $P_i^K := \bigcup_r P_{ir}$. Then $F(P_i^K) = q_i$, and H_i^K and G_i assign the same probability to every grid cell. Coupling the two beliefs by first drawing their common cell and then drawing within that cell gives $d_P(H_i^K, G_i) \leq \Delta$. ■

Proof of Proposition 6. Fix $K \geq 1$ and, as in Lemma 12, put $\Delta := (\bar{\theta} - \underline{\theta})/K$. Throughout the proof, C denotes a finite constant, independent of K and Δ , that may change from line to line. It may depend on the primitive bounds $\underline{\theta}, \bar{\theta}$ and may absorb constants such as C_0 from Lemma 11.

Let σ^* denote a consumer-optimal market segmentation and note that for σ^* -a.e. G , $p^{\min}(G) \in a^*(G)$. We assume throughout that $\underline{\theta} > 0$.³³

Assume K is large enough so that $\sqrt{\Delta} \in (0, 1/2)$, $2\Delta \leq \sqrt{\Delta}/4$, and $2\Delta \leq c_0$ (for small K we can construct an arbitrary equilibrium).

Step 1: Top-truncate the original segmentation. For each segment G in σ^* , form $G^{\sqrt{\Delta}}$. Since $(1 - \sqrt{\Delta})G^{\sqrt{\Delta}} \leq G$, the measure

$$F - \int (1 - \sqrt{\Delta})G^{\sqrt{\Delta}} d\sigma^*(G)$$

is positive and has mass $\sqrt{\Delta}$. Normalize it to obtain a residual segment. This gives a Bayes-plausible segmentation consisting of the main segments $\{G^{\sqrt{\Delta}}\}$, with total weight $1 - \sqrt{\Delta}$, plus a residual segment of weight $\sqrt{\Delta}$. By Lemma 11, the selected price $p^{\min}(G)$ is the unique optimal price under $G^{\sqrt{\Delta}}$. Since all payoffs are bounded by $\bar{\theta}$, this truncation and residual segment change the payoff profile by at most $C\sqrt{\Delta}$.

Step 2: Reduction to finite segmentation. The argument is analogous to the argument in Lemma 2. Cover the compact support of the segmentation created in Step 1 by finitely many d_P -balls of radius Δ , with centers chosen among the truncated segments $\{G^{\sqrt{\Delta}}\}$ themselves. Merge all top-truncated segments in the same cell; let $\{(\bar{q}_j, \bar{G}_j)\}_{j=1}^M$ be the resulting finite collection of main segments (that is, excluding the residual segment). For each j , choose the representative original segment G_j so that its truncation $G_j^{\sqrt{\Delta}}$ is the center of cell j 's ball. Since $\bar{G}_j$ is a barycenter of truncated segments lying in that ball, convexity of Lévy-Prokhorov balls places $\bar{G}_j$ in the same ball, so

$$d_P(\bar{G}_j, G_j^{\sqrt{\Delta}}) \leq \Delta.$$

The residual segment is kept as one additional segment.

By Lemma 11, every optimal price for $\bar{G}_j$ lies within $C\sqrt{\Delta}$ of $p^{\min}(G_j)$. Moreover, if a

³³The case $\underline{\theta} = 0$ reduces to this one: place $[0, \sqrt{\Delta}]$ in a segment of its own—a single interval—and apply the construction below to a consumer-optimal segmentation of the conditional market $F(\cdot \mid [\sqrt{\Delta}, \bar{\theta}])$, whose segments satisfy $p^{\min}(G) \geq \sqrt{\Delta}$. Isolating this bottom segment perturbs the consumer-optimal payoff by $O(\sqrt{\Delta})$, since it reallocates only consumers of value at most $\sqrt{\Delta}$ and the uniform-monopoly profit is insensitive to prices below $\sqrt{\Delta}$. Because C_0 does not depend on $\underline{\theta}$, the conditional market is still handled at rate $O(\sqrt{\Delta})$ —only $c_0 \asymp \sqrt{\Delta}$, which merely requires K large enough that $2\Delta \leq c_0$ —so the overall error remains $O(\sqrt{\Delta})$.

segment $G^{\sqrt{\Delta}}$ was merged into cell j , then $d_P(G^{\sqrt{\Delta}}, G_j^{\sqrt{\Delta}}) \leq \Delta$, so $p^{\min}(G)$ and $p^{\min}(G_j)$ are also within $C\sqrt{\Delta}$ of each other. Because $u_S(a, \theta)$ is 1-Lipschitz in a , the sender's payoff changes by at most $C\sqrt{\Delta}$. By Lemma 10, the receiver's payoff changes by at most $C\Delta$, since pooling the truncated segments of cell j moves U_R by at most $(1 + \bar{\theta}) d_P(\bar{G}_j, G^{\sqrt{\Delta}}) \leq C\Delta$. Hence the finite merging changes payoffs by at most $C\sqrt{\Delta}$.

Step 3: Replacing the finite segmentation by K -interval segmentations. Apply Lemma 12 to the finite segmentation from Step 2—the main segments together with the residual—and denote the resulting segmentation by σ^K . For each main segment,

$$d_P(H_j^K, \bar{G}_j) \leq \Delta,$$

and hence

$$d_P(H_j^K, G_j^{\sqrt{\Delta}}) \leq 2\Delta.$$

By Lemma 11, every optimal price under H_j^K lies within $C\sqrt{\Delta}$ of $p^{\min}(G_j)$. The receiver's payoff changes by at most $C\Delta$ by Lemma 10. For the sender, we bound the two changes separately. First, fix a price a . Since $\theta \mapsto u_S(a, \theta)$ is 1-Lipschitz, by Lemma 12, replacing $\bar{G}_j$ by H_j^K at this fixed price changes the sender's payoff by at most $C\Delta$. Second, by Lemma 11, the optimal price under H_j^K is within $C\sqrt{\Delta}$ of $p^{\min}(G_j)$, as is the optimal price under $\bar{G}_j$; since u_S is 1-Lipschitz in a , this changes the sender's payoff by at most $C\sqrt{\Delta}$. Thus, the sender's payoff changes by at most $C\sqrt{\Delta}$. The residual segment has mass at most $\sqrt{\Delta}$, so its total contribution is bounded by $C\sqrt{\Delta}$.

Combining the three steps delivers $|u_S^K - u_S^*| + |u_R^K - u_R^*| \leq C\sqrt{\Delta}$. Since $\Delta = (\bar{\theta} - \underline{\theta})/K$, this is $O(K^{-1/2})$.

It remains to support these payoffs as an equilibrium. The sets $\{P_j^K\}$ partition Θ up to endpoints, so the segmentation σ^K constructed above is finite partitional, and each segment is supported on a union of at most K intervals. Since monopoly pricing satisfies Assumptions 1 and 2, Lemma 1 yields an equilibrium of the disclosure game with payoffs (u_S^K, u_R^K) in which every on-path message is a union of at most K intervals. ■

Proof of Proposition 7. Fix an achievable profile (u_S^*, u_R^*) . The game augmented with payoff-irrelevant labels is the disclosure game with type space $\tilde{\Theta} := \Theta \times [0, 1]$, prior $\tilde{F} := F \otimes \lambda$ (λ Lebesgue), message set $\tilde{\mathcal{C}}$, and payoffs that depend on (θ, x) only through θ ; a belief over $\tilde{\Theta}$ thus matters only through its θ -marginal.

Because Θ is finite, by the Carathéodory theorem, (u_S^*, u_R^*) is generated by a finite segmentation $\{(q_i, G_i)\}_{i=1}^n$ of F , where G_i is the i -th segment and q_i is the probability assigned to this segment, with best responses $\{a_i\}_{i=1}^n$, where $a_i \in a^*(G_i)$ for each $i = 1, \dots, n$. For each θ , Bayes plausibility makes $\boldsymbol{\pi}(\theta) := (\pi_1(\theta), \dots, \pi_n(\theta))$, where $\pi_i(\theta) := q_i G_i(\theta)/F(\theta)$,

a probability vector. Partition $[0, 1]$ into intervals $\{I_i(\theta)\}_{i=1}^n$, each of length $\pi_i(\theta)$, and set $s_i := \bigcup_{\theta \in \Theta} \{\theta\} \times I_i(\theta)$. Then $\{s_i\}_{i=1}^n$ partitions $\tilde{\Theta}$, with $\tilde{F}(s_i) = q_i$ and $\tilde{F}(\cdot | s_i)$ having θ -marginal G_i for each i . Let $\tilde{\sigma} := \{(q_i, \tilde{F}(\cdot | s_i))\}_{i=1}^n$; as payoffs depend only on θ , $a_i \in a^*(\tilde{F}(\cdot | s_i))$ for each i , so $\tilde{\sigma}$ yields (u_S^*, u_R^*) .

The segmentation $\tilde{\sigma}$ is finite partitional and [Assumptions 1](#) and [2](#) hold in the augmented game. [Lemma 1](#) therefore yields an equilibrium with payoffs (u_S^*, u_R^*) . ■